GAS CHROMATOGRAPHY

CLARUS 600/680/690 GC

Installation Guide

PerkinElmer®

Release History

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09936779	E	August 2017

Any comments about the documentation for this product should be addressed to:

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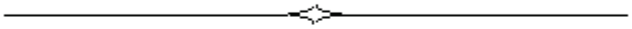
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Introduction 1



About This Manual

This quick installation guide is divided into following chapters:

Chapter 1 - Introduction

This chapter contains a brief introduction of the instrument, the conventions and warnings used in the manual.

Chapter 2 - Safety Practices

Important safety information for the Clarus GC is provided in this chapter.

Chapter 3 - Preparing Your Laboratory

The preparation of your laboratory needed for the Clarus GC laboratory requirements for the instrument are reviewed in this chapter.

Chapter 4 - Installing the Clarus GC

Information on connecting the Clarus GC is provided.

Chapter 5 - Connecting the Gases

Information on connecting the gases and electrical supply to the system is provided.

Chapter 6 - Connecting the Accessories

Information on installing accessories such as the NPD Bead Assembly, recorder, integrator and the printer is provided.

Chapter 7 - PPC Restrictor Information

PPC restrictor information is provided in this chapter.

Conventions Used in this Manual

- Normal text is used to provide information and instructions.
- **Bold** text refers to button or tab page that is displayed on the touch screen.
- All eight digit numbers are PerkinElmer part numbers unless stated otherwise.

About Notes, Cautions and Warnings

Three terms, in the following standard formats, are also used to highlight special circumstances and warnings.

NOTE: *A note indicates additional, significant information that is provided with some procedures.*

CAUTION	*We use the term CAUTION to inform you about situations that could result in **serious damage to the instrument** or other equipment. Details about these circumstances are in a box like this one.*

 **WARNING**	*We use the term WARNING to inform you about situations that could result in **personal injury** to yourself or other persons. Details about these circumstances are in a box like this one.*

Customer Service

PerkinElmer, Inc.
710 Bridgeport Avenue
Shelton, CT 06484-4794 U.S.A

Tel: 1 (800) 762-4000 or (203) 762-4000
Internet: http://www.perkinelmer.com

Electromagnetic Compatibility (EMC)

This product has been designed and manufactured, having regard to the state of the art, to ensure that:

(a) the electromagnetic disturbance generated does not exceed the level above which radio and telecommunications equipment or other equipment cannot operate as intended;

(b) it has a level of immunity to the electromagnetic disturbance to be expected in its intended use which allows it to operate without unacceptable degradation of its intended use.

United States (FCC)

United States (FCC) This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a business/industrial/commercial environment is likely to cause harmful interference in which the user will be required to correct the interference at your own expense. Changes or modifications not expressly approved by the manufacturer could void your authority to operate the equipment in compliance with FCC rules.

NOTE: *Changes or modifications not expressly approved by PerkinElmer could cause the instrument to violate FCC (U.S. Federal Communications Commission) emission regulations, and because of this violation could void the user's authority to operate this equipment. Use of this product in residential areas may cause interference and such use should be avoided unless special measures are taken by the user to restrict emissions to a level that allows the reception of broadcast emissions.*

European Union

All information concerning EMC standards will be in the Declaration of Conformity and these standards will change as the European Union adds new requirements.

European Union Industrial Environment

The 230 V/50 Hz. Clarus GC manufactured for use in the European Union is intended for the industrial environment. The instrument is to be connected to a mains power network supplied from a high or medium-voltage transformer dedicated for the supply of an installation feeding a manufacturing or similar plant.

Industrial environments are characterized by the existence of one or more of the following conditions:

-industrial, scientific and medical (ISM) apparatus are present

-heavy inductive or capacitive loads are frequently switched

-currents and associated magnetic fields are high

These are the major contributors to the industrial electromagnetic environment and as such distinguish the industrial from other environments. The instrument is not intended for connection to a public mains network supplying residential, commercial and light-industrial locations.

South Korea

This device complies with MSIP (Ministry 0f Science, ICT, and Future Planning) EMC Registration requirements. This instrument is registered as a Class A instrument for buiness use only. Product seller and user should notice that this equipment is not for house hold use.

A급 기기 (업무용 방송통신기자재)
이 기기는 업무용(A급) 전자파적합기기로서 판
매자 또는 사용자는 이 점을 주의하시기 바라
며, 가정외의 지역에서 사용하는 것을 목적으
로 합니다.

Susceptibility to RF Interference

With the exception of the Flame Ionization Detector (FID), a RF field strength of 10 V/m between 80 MHz. and 1000 MHz. with 80% modulation at 1 kHz. may cause a deflection on the chromatographic detector baseline that exceeds its normal pattern. This implies that if a transmitting device, such as a walkie-talkie carried by a security guard, is used near the detector, a spike or peak on the chromatographic baseline may occur. If you are concerned that such an event may occur, PerkinElmer recommends that walkie-talkie restriction notices be posted in the vicinity. Cell phones, beepers, and other similar devices operate in a much higher frequency range and do not cause interference.

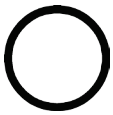
Warning Labels on the Instrument



Alternating current.



Protective conductor terminal.



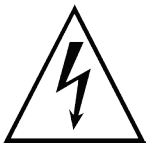
Off position of the main power source.



On position of the main power source.



Warning, hot surface



Warning, risk of electric shock.



Caution, risk of danger.
Documentation must be consulted to determine the nature of the potential hazard and any actions which have to be taken.



This unit contains protective circuitry. Contact PerkinElmer qualified service personnel before performing any AC line tests.



Caution

If the instrument top is opened, the alignment of the Autosampler tower must be verified. Refer to your Clarus Manual.

(The CAUTION above is only visible if detector compartment door is open.)



Warning

Disconnect AC power cord from the outlet before removing any covers or parts. Do not operate the instrument with any cover or parts removed.

(The WARNING above is only visible if detector compartment door is open.)



Warning

Grounding circuit continuity is vital for safe operation of equipment. Never operate equipment with grounding conductor disconnected.



Warning

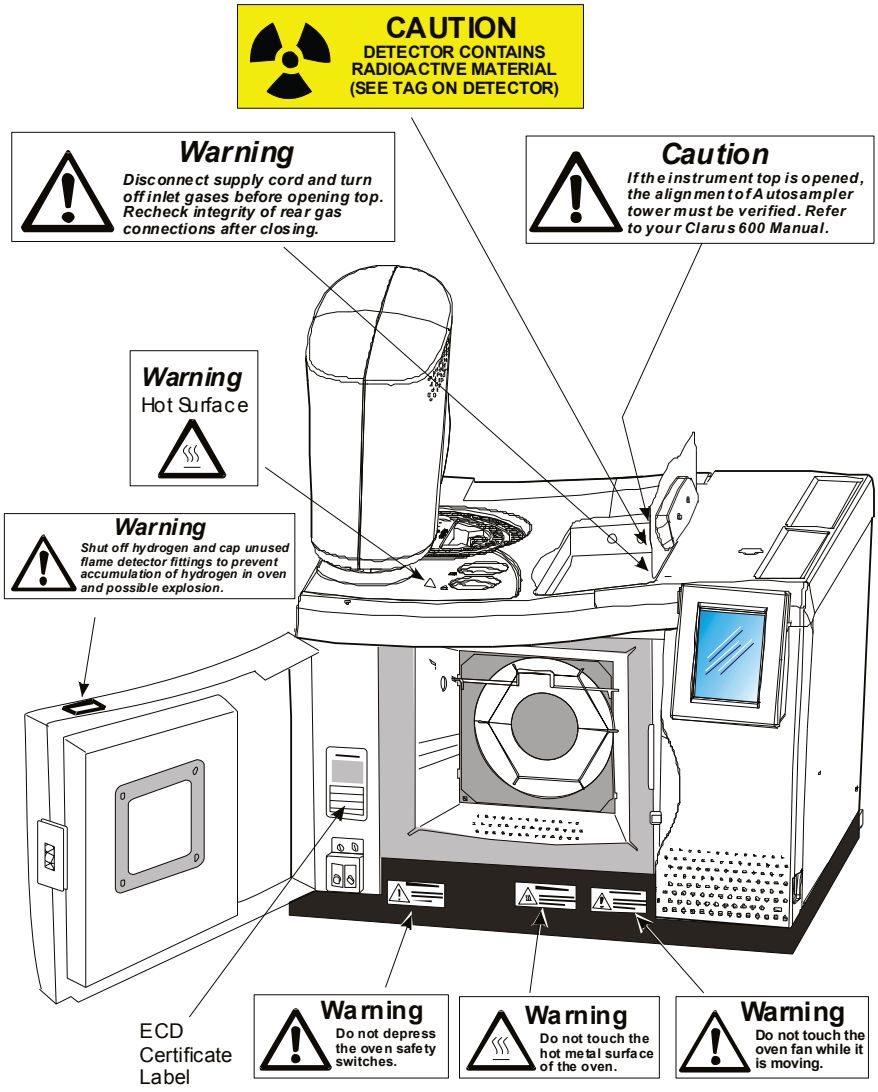
Shut off hydrogen and cap unused flame detector fittings to prevent accumulation of hydrogen in oven and possible explosion.

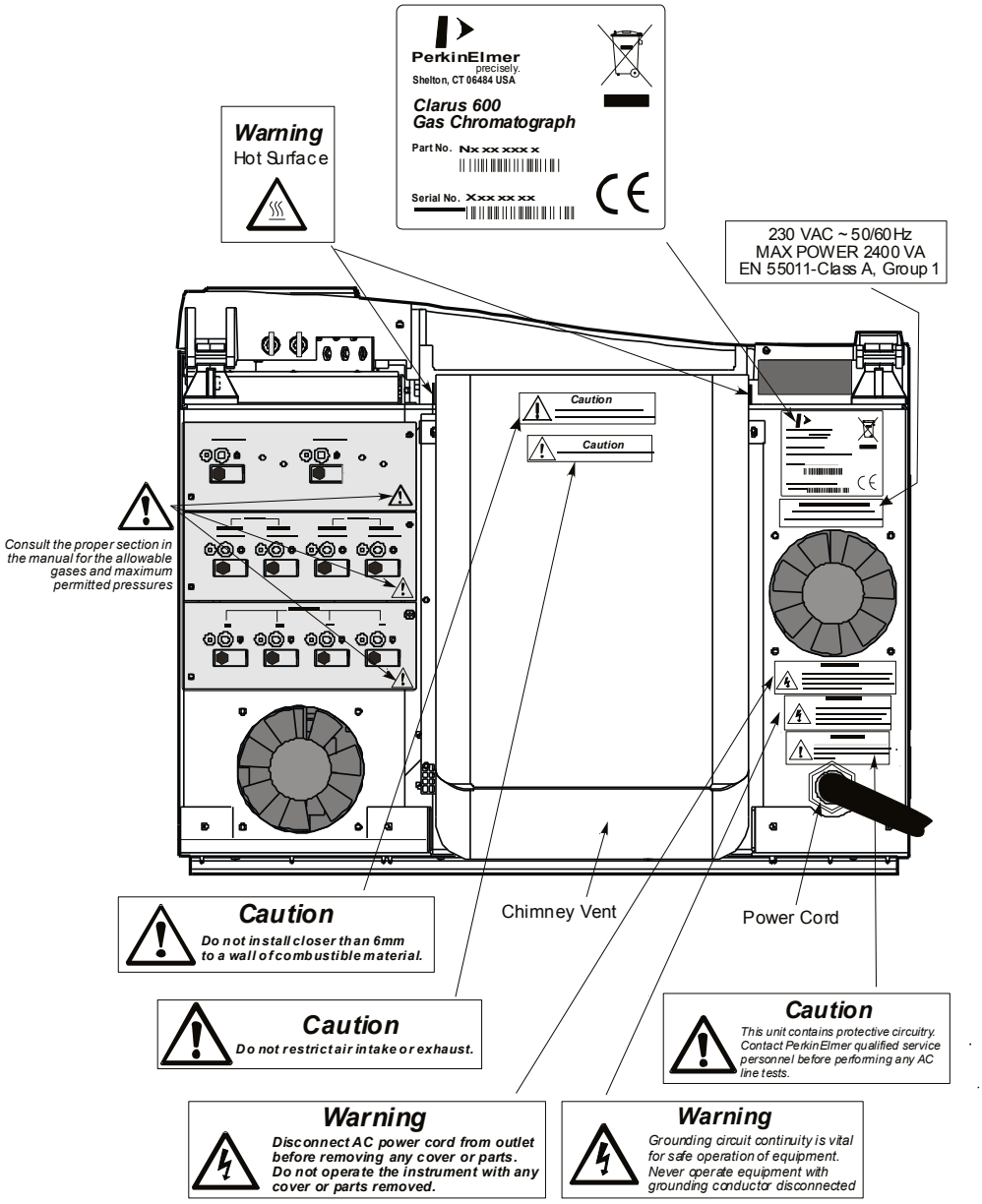
(The WARNING above is only visible if detector compartment door is open.)

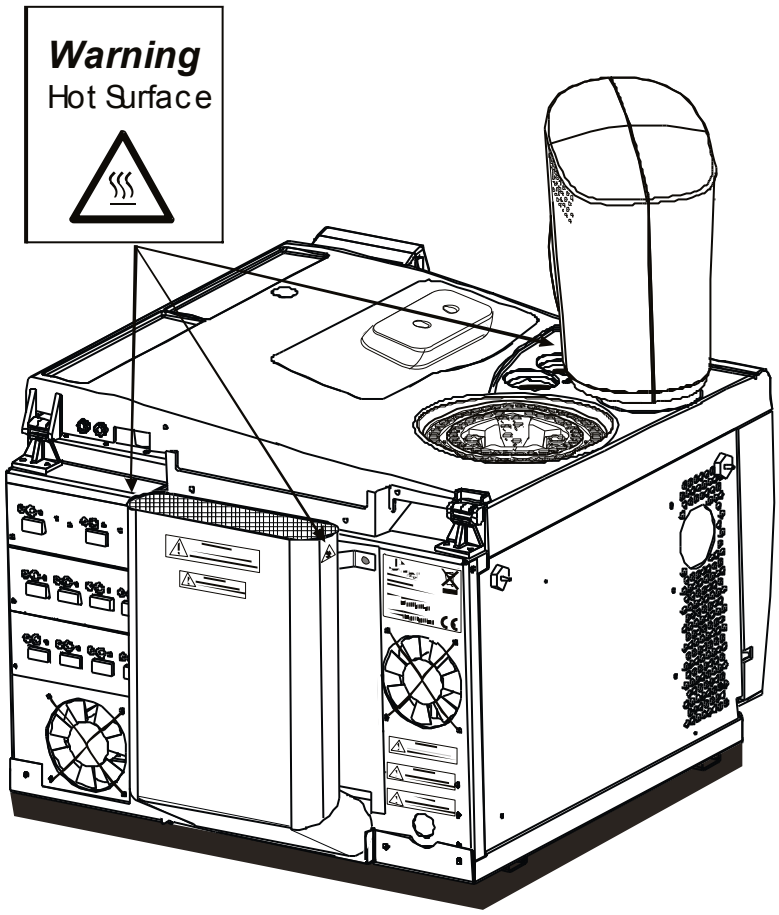
For ECD Instruments only the following certificate appears:

CERTIFICATE **^{63} Ni Electron Capture Detector** **_____ N6100134 ____N61000063** **Save for proof of tests.** **The attached source complies with the requirements of ANSI N542 (ISO 2919) Classification C42211. This source has been leak tested and is certified to have less than 0.005 µCi, 180 Bq of removable contamination.**			
Serial Number			
Assembly Number			
Isotope 63 Ni Activity: 15 mCi			
Pressurization test at 30 psi N_2: **Passed:**			
Initial Test Performed by:			
Wipe Test Date			

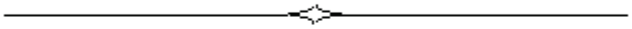
Label locations on instrument




PerkinElmer
precisely.
Shelton, CT 06484 USA
Clarus 600
Gas Chromatograph
Part No. Nxxxxxxx
Serial No. Xxxxxx
Warning
Hot Surface
230 VAC ~ 50/60 Hz
MAX POWER 2400 VA
EN 55011-Class A, Group 1
Caution
Caution
Consult the proper section in the manual for the allowable gases and maximum permitted pressures
Caution
Do not install closer than 6mm to a wall of combustible material.
Chimney Vent
Power Cord
Caution
Do not restrict air intake or exhaust.
Caution
This unit contains protective circuitry. Contact PerkinElmer qualified service personnel before performing any AC line tests.
Warning
Disconnect AC power cord from outlet before removing any cover or parts. Do not operate the instrument with any cover or parts removed.
Warning
Grounding circuit continuity is vital for safe operation of equipment. Never operate equipment with grounding conductor disconnected


Warning
Hot Surface

Safety Practices 2



Chapter Overview

This chapter describes the general safety practices and precautions that must be observed when operating the Clarus 600/680/690 GC.

This equipment requires no specified inspection or preventive maintenance to ensure the continuous functioning of its safety features.

This advice is intended to supplement, not supersede, the normal safety codes in the user's country. It is also a supplement to the PerkinElmer standard Safety and Health Policy. The information provided does not cover every safety procedure that should be practiced. Ultimately, maintenance of a safe laboratory environment is the responsibility of the analyst and the analyst's organization.

Please consult all manuals supplied with the Clarus 600/680/690 GC and accessories before you start working with the instrument. Carefully read the safety information in this chapter and in the other manuals supplied. When setting up the instrument or performing analyses or maintenance procedures, strictly follow the instructions provided. The Clarus GC should be used in accordance with the instructions provided in this manual. If used otherwise, the protection provided by the instrument may be impaired.

Generic Warnings

Before installing or operating the Clarus 600/680/690 GC, read the following information concerning hazards and potential hazards. You should ensure that anyone involved with installation and/or operation of the GC is knowledgeable in both general safety practices for the laboratory and safety practices for the GC. Get advice from your safety engineer, industrial hygienist, environmental engineer, or safety manager before you install or use this instrument.

Heated Zones

Heated zones should be treated with caution, for example, injector caps and detectors. Avoid physical contact with the injector caps. The detector cover may get hot, especially if flame ionization detectors are operated at high temperatures. As a general rule, allow heated zones to cool before attempting to work in the oven, injector, or detector areas.

CAUTION

THERMAL RUNAWAY PROTECTION: *The Clarus 600/680/690 GC software shuts down the instrument if any heated zone exceeds 470 °C. Should this occur, the following error message is displayed*:

INSTRUMENT SHUTDOWN
xxx THERM RUNAWAY where xxx is the heated zone

Call your PerkinElmer Representative.

Instrument shutdown also occurs if there is a PRT (Platinum Resistance Thermometer) or MPU (Micro Processor Unit) failure. In these cases the following error message is displayed:

INSTRUMENT SHUTDOWN
xxx PRT ERROR where xxx is the failed zone.

Call your PerkinElmer Representative.

Precautions



WARNING

Be sure that all instrument operators read and understand the precautions listed below. It is advisable to post a copy of the precautions near or on the instrument shelf.

CAUTION

The protection provided by this equipment may be impaired if the equipment is used in a manner not specified by PerkinElmer.



WARNING

To reduce the chance of electrical shock, do not remove covers that require tool access. No user serviceable parts are inside. Refer servicing to PerkinElmer or other qualified service personnel.



WARNING

To avoid electrical shock, disconnect power cord before servicing.

CAUTION

*Do **not** restrict air intake or exhaust.*

The following precautions must be observed when using the Clarus 600/680/690 GC:

- PerkinElmer provides a number of safety interlocks on its instruments that monitor such things as door locks, gas pressures, and the temperatures of various components. Do **not** attempt to defeat these interlocks; you may compromise your own or someone else's safety.

- Be sure that the power line voltage of the Clarus 600/680/690 GC corresponds to the voltage used in your laboratory.

- Never remove the side panels of the Clarus 600/680/690 GC without shutting down the instrument and disconnecting the instrument power cord from line power.

- Do **not** immerse the purge gas exit line in a liquid as the liquid may be drawn back into the sample holder.

- Only high quality purge gases should be used with the Clarus 600/680/690 GC. Minimum purity of 99.995% is recommended. A high quality filter-dryer accessory is recommended for the removal of any moisture from the purge gases.

Environmental Conditions

Operating Conditions

CAUTION

The Clarus 600/680/690 GC is designed for indoor use only.



WARNING

*The Clarus 600/680/690 GC is **not** designed for operation in an explosive environment.*

Pollution Degree

The Clarus GC will operate safely in environments that contain nonconductive foreign matter up to Pollution Degree 2 in EN/IEC 61010-1.



WARNING

Dispose of waste in accordance with the regulations applicable to your locality, state and/or country.

Clarus GC Touch Screen

For optimum performance, the Clarus GC's touch screen may require periodic re-calibration. The interval between re-calibration may be affected by exposure to combined heat and humidity conditions (ambient conditions between 30 °C / 50% RH and 35 °C / 80% RH).

Normal Operating Conditions

- Indoors,
- Temperature: +10 °C to +35 °C (+50 °F to +95 °F),
- Relative humidity: 20% to 80%, without condensation,
- Altitude: in the range -400 m to 2,000 m (sea level to 6,562 feet)

Safe Operating Conditions

Do **not** operate in a Cold Room or a refrigerated area.

- Indoors.
- Temperature: +5°C to +40°C (+41°F to +104°F)
- Relative humidity: 20% to 80%, without condensation
- Altitude: in the range -400 m to 2,000 m (sea level to 6,562 feet)

Storage Conditions

The Clarus GC may be stored under the following conditions:

- Ambient temperature is -20 °C to +60 °C (-4 to 140 °F)
- Ambient relative humidity is 20 to 80%, non-condensing
- Altitude is in the range -400 m to 12 000 m (sea level to 39 370 feet).

Cleaning the Instrument

Exterior surfaces may be cleaned with a soft cloth, dampened with a mild detergent and water solution. Do not use abrasive cleaners or solvents.

Decontamination and Cleaning

Before using any cleaning or decontamination methods except those specified by PerkinElmer, users should check with PerkinElmer that the proposed method will not damage the equipment.

Decontamination

Customers wishing to return instrumentation and/or associated materials to PerkinElmer for repair, maintenance, warranty or trade-in purposes are advised that all returned goods must be certified as clean and free from contamination.

The customer's responsible body is required to follow the "Equipment Decontamination Procedure" and complete the "Certificate of Decontamination". These documents are available on the PerkinElmer public website:

http://www.perkinelmer.com/Content/technicalinfo/dts_instrumentdeconprocedure.pdf

Customer Care USA:	**1-800-762-4000**	**(inside the USA)**
(8:30 a.m. – 7 p.m. EST)	**(+1) 203-925-4602**	**(outside the USA)**
Customer Care Canada:	**800-561-4646**	
Customer Care EU:	**0800 40 858**	**(Brussels)**
	0800 90 66 42	**(Monza)**

If you are located outside of the United States or Canada, please call your local PerkinElmer sales office for more information.

General Laboratory Safety

Your laboratory should have all equipment ordinarily required for the safety of individuals working with chemicals (fire extinguishers, first-aid equipment, safety shower and eye-wash fountain, spill cleanup equipment, etc.).

Electrical Safety

The Clarus 600/680/690 GC contains high voltage. To prevent the risk of shock, unplug the line cord from the AC outlet and wait at least one minute before opening or removing any instrument panels.

The instrument has been designed to protect the operator from potential electrical hazards. This section describes some recommended electrical safety practices.

CAUTION *This unit contains protective circuitry. Contact PerkinElmer Service before performing any AC line tests.*



WARNING *Connect the GC to an AC line power outlet that has a protective ground connection. To ensure satisfactory and safe operation of the GC, it is essential that the protective ground conductor (the green/yellow lead) of the line power cord is connected to a true electrical ground. Any interruption of the protective ground conductor, inside or outside the GC, or disconnection of the protective ground terminal may impair the protection provided by the GC.*



WARNING ***Do not operate the GC with any covers or parts removed.***



WARNING *To avoid electrical shock, disconnect the power cord from the AC outlet before servicing. Servicing on the GC is to be performed only by a PerkinElmer service representative or similarly trained and authorized person.*



WARNING

*Do **not** attempt to make adjustments, replacements or repairs to this GC except as described in the user documentation.*

CAUTION

To ensure adequate cooling of the instrument electronics, do not obstruct the gap at the base of the GC, and leave at least a 6-inch clearance between instruments.

Ensure that the power cord is correctly wired and that the ground leads of all electrical units (for example, recorders, integrators) are connected together via the circuit ground to earth. Use only three-prong outlets with common earth ground connections.

For customers in the United Kingdom, the power plug provided with the Clarus 600/680/690 GC contains a 13 Amp fuse mounted within the plug body. This fuse complies with standard BS1362 and has a breaking capacity of 6000 Amps at 250 VAC. Any replacement fuse must be of the same type and amperage rating.

Servicing of incoming AC line components in your laboratory should be performed only by a licensed electrician.



WARNING

Lethal voltages are present at certain areas within the instrument. Installation and internal maintenance of the instrument should only be performed by a PerkinElmer service engineer or similarly authorized and trained person. *When the instrument is connected to line power, opening the instrument covers is likely to expose live parts. Even when the power switch is off, high voltages can still be present. Capacitors inside the instrument may still be charged even if the instrument has been disconnected from all voltage sources.*

The instrument must be correctly connected to a suitable electrical supply. The supply must have a correctly installed protective conductor (earth ground) and must be installed or checked by a qualified electrician before connecting the instrument.



WARNING

Any interruption of the protective conductor (earth ground) inside or outside the instrument or disconnection of the protective conductor terminal is likely to make the instrument dangerous. Intentional interruption is prohibited.

When working with the instrument:

- Do **not** position the equipment so that it is difficult to operate the AC power switch.
- Disconnect the instrument from all voltage sources before opening it for any adjustment, replacement, maintenance, or repair. If afterwards, the opened instrument must be operated for further adjustment, maintenance, or repair, this must only be done by a PerkinElmer Service engineer.
- Whenever it is possible that the instrument is no longer electrically safe for use, make the instrument inoperative and secure it against any unauthorized or unintentional operation. The electrical safety of the instrument is likely to be impaired if, for example, the instrument shows visible damage, has been subjected to prolonged storage under unfavorable conditions, or has been subjected to severe stress during transportation.

Moving the Clarus 600/680/690 GC



WARNING

Risk of injury through lifting. The instrument weighs 64.5kg (142 lb.). Improper lifting can cause painful and sometimes permanent back injury.
Use proper lifting procedures and/or mechanical lifting aids to lift or move the instrument. If the instrument must be moved, we recommend that at least two people carefully lift the instrument in order to move it.

ECD Radioactive Hazards



WARNING

To assure that removable radioactive contamination on the external parts of the ECD remains at a safe level, the United States Nuclear Regulatory Commission requires that:

- *Wipe test the ECD at least once every six months.*
- *A record of the results must be maintained for NRC inspection.*

United States Government Regulations for ECDs

NOTE: *To repair an Electron Capture Detector cell requires a specific license issued by the U.S. Nuclear Regulatory Commission (NRC) and/or in some states by the equivalent state agency. For further information on obtaining a license, contact the Customer Service Department at PerkinElmer, Shelton, Connecticut, or the NRC Material Branch, Office of Nuclear Materials, Safety and Safeguards, Washington, DC 20555.*

All USNRC regulations can be obtained through the internet at www.nrc.gov/reading-rm/

NOTE: *These instructions are for ECD cell purchasers who are not specifically licensed to handle radioactive materials.*

The Clarus 600/680/690 GC Electron Capture Detector model (P/N N6100063) contains a maximum of 15 mCi of Nickel 63 (Ni 63), a radioactive material. Your possession and use of this detector is governed by 10 C.F.R. Section 31.5 which is reproduced in Appendix I. Under the provisions of that regulation you are deemed a "General Licensee."

Your possession and use of the detector cell may also be regulated by the state where you are located. The requirements of state regulatory agencies are substantially similar to those contained in NRC regulation 10 C.F.R. Section 31.5, but they may differ in some respects. It is suggested that you procure a copy of the regulations of your particular state. (Supplement 2 in Appendix I contains a list of the "Agreement States" which have been granted authority by the U.S. Nuclear Regulatory Commission to regulate the possession and use of radioactive material.)

It is required that you be familiar with regulation 10 C.F.R. Section 31.5 (Appendix 1 in the Clarus Hardware Guide). Following are summaries of its requirements.

Labels

Do not remove any of the labels attached to the ECD cell or any of the labels attached to your Clarus Gas Chromatograph that refer to the ECD cell. Follow all instructions and abide by all precautions provided by the labels and in user instruction manuals referred to by the labels.

Leak Testing

You are obligated under U.S. federal and state regulations to make certain that the ECD cell is wipe-tested for leakage of radioactive materials at intervals of no longer than six months, and that the analysis of these wipe tests is conducted by a person specifically licensed to do so, either by the U.S. Nuclear Regulatory Commission or by an Agreement State. The analyses can be performed by the firm listed below:

National Leak Test Center
P.O. Box 486
North Tonawanda, New York 14120

ECD Cell Failure or Damage

If a leak test detects more than 0.005 μCi (microcurie) of removable radioactive material on the surface of an ECD cell, or if the cell itself is damaged in such a way as to indicate that it may no longer adequately shield the radioactive material inside, you must immediately suspend operation of your chromatograph until the cell has been repaired or disposed of by ***a person specifically licensed to do so***. Any such incident must be reported by you to the Regional Office, Inspection and Enforcement, U.S. Nuclear Regulatory Commission.

Reporting Radiation Incidents, Theft or Loss

Please read Regulation 10 C.F.R. Section 20.2201 and 20.2202. These describe your duties should the radioactive material (Ni 63) in the ECD cell be lost, stolen, or released, or should any person be exposed to radiation.

Other ECD Requirements

Regulation 10 C.F.R. Section 31.5 (see Appendix I) does not permit you to abandon the ECD cell or export it. It may not be transferred except to a person specifically licensed to receive it. Within thirty days of such a transfer, you must report to the Director of Nuclear Material Safety and Safeguards, U.S. Nuclear Regulatory Commission, Washington, D.C. 20555, the name and address of the transferee. However, no report is needed to transfer a defective ECD cell to PerkinElmer in order to obtain a replacement.

You may transfer the ECD cell to another general licensee, like yourself, only when it remains at the same location to which it was shipped by PerkinElmer. Give the transferee a copy of these instructions and the regulations in Appendix I, and report to the commission as required in Regulation C.F.R. Section 31.5.



WARNING

NEVER DISMANTLE THE ECD CELL!!

You can remove the ECD cell from the GC for repair by a person specifically licensed to do so..

United Kingdom Regulations

In the U.K., registration is required under the Radioactive Substances Act of 1960, for anyone keeping or using radioactive materials. Application should be made to any one of the following governing bodies:

ENGLAND Department of the Environment
Queen Anne's Chambers
Tothill Street
London, SW1H 9J4

SCOTLAND Scottish Development Department
21 Hill Street
Edinburgh, EH2 3J4

WALES Welsh Office
Cathay's Park
Cardiff, CF1 3NG

NORTHERN IRELAND Ministry of Development
Parliament Building
Storemont
Belfast, Northern Ireland

Safe Handling of Gases

When using hydrogen, either as the combustion gas for a flame ionization detector or as a carrier gas, special care must be taken to avoid buildup of explosive hydrogen/air mixtures. Ensure that all hydrogen line couplings are leak-free and do not allow hydrogen to vent within the oven.



WARNING

Warning: Compressed Gases
High pressure gas cylinders can be dangerous if mishandled or misused. Always handle gas cylinders with caution and observe your local regulations for the safe handling of gas cylinders

Ventilation

Adequate ventilation must be provided. When analyzing hazardous compounds, such as pesticides, it may be necessary to arrange for venting of detector effluent into a fume hood.

Using Hydrogen



WARNING

Flame Ionization Detectors (FID) uses hydrogen as fuel. If the hydrogen is turned on without a column attached to the injector and detector fittings inside the oven, hydrogen could diffuse into the oven creating the possibility of an explosion. To avoid possible injury, ***DO NOT TURN ON THE HYDROGEN UNLESS A COLUMN IS ATTACHED AND ALL JOINTS HAVE BEEN LEAK TESTED.***

Before disconnecting a column, make certain that the hydrogen has been turned ***OFF.***

If two FIDs are installed and only one has a column attached to it, make certain that you cap off the unused detector inlet fitting with a 1/8-inch stainless steel plug (Part No. N9300061).



WARNING

Contact the gas supplier for a material safety data sheet (MSDS) containing detailed information on the potential hazards associated with the gas. *Carefully use, store, and handle compressed gases in cylinders. Gas cylinders can be hazardous if they are mishandled.*

NOTE: *The permanent installation of gas supplies is the responsibility of the user and should conform to local safety and building codes.*

Consult the following references for more detailed information and additional guidelines about gas cyclinders.

- Compressed Gas Association (USA), "Safe Handling of Compressed Gases in Containers," pamphlet no. P-1, 1984.
- Compressed Gas Association (USA), "The Inert Gases – Argon, Nitrogen and Helium," pamphlet no. P-9, 1992.

Identification of Gas Cylinders

- Legibly mark cylinders to identify their contents. Use the chemical name or commercially accepted name for the gas.

Storing Gas Cylinders

Review the following precautions with the customer to ensure the safe use and storage of gas cylinders.

- Cylinders should be stored in accordance with the regulations and standards applicable to the customer's locality, state, and country.
- When cylinders are stored indoors in storage rooms, the storage room should be well ventilated and dry. Ensure that the ventilation is adequate to prevent the formation of dangerous accumulations of gas. This is particularly important in small or confined areas.
- Do not store cylinders near elevators, gangways, or in locations where heavy moving objects may strike or fall against them.
- Use and store cylinders away from exits and exit routes.
- Locate cylinders away from heat sources, including heat lamps. Compressed gas cylinders should not be subjected to temperatures above 52 °C (126 °F).
- It is recommended that gas cylinders be stored and placed outside the laboratory and connected to the instrument through copper lines.

Handling of Gas Cylinders

- Do not allow ignition sources in the storage area and keep cylinders away from readily ignitable substances such as gasoline or waste, or combustibles in bulk, including oil.
- Store cylinders standing upright, fastened securely to an immovable bulkhead or permanent wall.
- When storing cylinders outdoors, they should be stored above ground on a suitable floor and protected against temperature extremes (including the direct rays of the sun).
- Arrange gas hoses where they will not be damaged or stepped on and where things will not be dropped on them.
- Take care not to kink or stress the gas lines. For safety, cylinders should be firmly clamped in position.
- If it becomes necessary to move the cylinders, do so with a suitable hand truck after insuring that the container cap is secured and the cylinder is properly fastened to the hand truck.
- Use only regulators, tubing and hose connectors approved by an appropriate regulatory agency.
- Do not refill cylinders.
- Check the condition of pipes, hoses and connectors regularly. Perform gas leak tests at all joints and seals of the gas system regularly, using an approved gas leak detection system.
- When the equipment is turned off, close all gas cylinder valves tightly at the cylinder. Bleed the remainder of the line before turning the exhaust vent off.

Hazardous Chemicals

Before using samples, you should be thoroughly familiar with all hazards and safe handling practices. Observe the manufacturer's recommendations for use, storage and disposal. These recommendations are normally provided in the Material Safety Data Sheets (MSDS) supplied with the solvents.

Be aware that the chemicals that you use in conjunction with the GC may be hazardous. **DO NOT** store, handle, or work with any chemicals or hazardous materials unless you have received appropriate safety training and have read and understood all related Material Safety Data Sheets (MSDS). MSDS's provide information on physical characteristics, precautions, first aid, spill clean up and disposal procedures. Familiarize yourself with the information and precautions contained in these documents before attempting to store, use or dispose of the reagents. Comply with all federal, state, and local laws related to chemical storage, handling, and disposal.

You must work under a suitable hood when handling and mixing certain chemicals. The room in which you work must have proper ventilation and a waste collection system. Always wear appropriate safety attire (full-length laboratory coat, protective glasses, gloves, etc.), as indicated on Material Safety Data Sheets.



WARNING

Some chemicals used with this GC may be hazardous or may become hazardous after completion of an analysis. The responsible body (for example, the Lab Manager) must take the necessary precautions to ensure that the GC operators and the surrounding workplace are not exposed to hazardous levels of toxic substances (chemical or biological) as defined in the applicable Material Safety Data Sheets (MSDS) or OSHA, ACGIH, or COSHH documents. Venting for fumes and disposal of waste must be in accordance with all national, state and local health and safety regulations and laws.

Definitions in Warning for Hazardous Chemicals

Responsible body. "Individual or group responsible for the use and maintenance of equipment, and for ensuring that operators are adequately trained." [per IEC 61010-1].

Operator. "Person operating equipment for its intended purpose." [per IEC 61010-1].

OSHA: Occupational Safety and Health Administration (United States)

ACGIH: American Conference of Governmental Industrial Hygienists

COSHH: Control of Substances Hazardous to Health (United Kingdom)

WEEE Instructions for PerkinElmer Products



or



A label with a crossed-out wheeled bin symbol and a rectangular bar indicates that the product is covered by the Waste Electrical and Electronic Equipment (WEEE) Directive and is not to be disposed of as unsorted municipal waste. Any products marked with this symbol must be collected separately, according to the regulatory guidelines in your area.

The objectives of this program are to preserve, protect and improve the quality of the environment, protect human health, and utilize natural resources prudently and rationally. Specific treatment of WEEE is indispensable in order to avoid the dispersion of pollutants into the recycled material or waste stream. Such treatment is the most effective means of protecting the customer's environment.

Requirements for waste collection, reuse, recycling, and recovery programs vary by regulatory authority at your location. Contact your local responsible body (e.g., your laboratory manager) or authorized representative for information regarding applicable disposal regulations. Contact PerkinElmer at the web site listed below for information specific to PerkinElmer products.

Web address:

http://www.perkinelmer.com/WEEE

For Customer Care telephone numbers select "Contact us" on the web page.

Products from other manufacturers may also form a part of your PerkinElmer system. These other producers are directly responsible for the collection and processing of their own waste products under the terms of the WEEE Directive. Please contact these producers directly before discarding any of their products.

Consult the PerkinElmer web site (above) for producer names and web addresses.



Preparing Your Laboratory 3



Chapter Overview

The items shown in the following checklist need to be considered when preparing the laboratory for the instrument.

- Pre-Installation Checklist
- Environmental Conditions
- Exhaust Vent Requirements (if required)
- Laboratory Space Requirements
- Cooling Water Requirements (if required)
- Electrical Requirements
- Pneumatic Requirements (if required)

Critical Checks Prior to Installation



WARNING

Measure the Line voltage at the Customer's site before starting the Install Process.

Measured Line Voltage =_____

Pre-Installation Checklist

Model: _____ Date: _____

Customer: _____

SPO# _____

Installation Requirements	OK	Needs Prior to Installation
Measure Line Voltage to determine heater Requirements		
Determine the Proper electrical connections		
Determine Proper Heater installation in the oven.		
Heat Venting from the oven.		
Lab Space Requirements Peripherals		
Safety Requirements		
Gas Requirements		
Sample Preparation (Customer Responsibility)		
Computer Configuration		
Miscellaneous		

Connect the Electrical Supply

This section describes how to connect the Clarus 600/680/690 GC to AC line power both in and outside the U.S.

WARNING

Only a PerkinElmer Service representative or similarly trained and authorized personnel should ever change or replace the plug for the Clarus 600/680/690 GC.

Connecting the Clarus 600/680/690 GC to Line Power in the U.S.

120 Volt Version Slow Heating Rates

To connect the Clarus 600/680/690 GC to line power in the U.S., simply plug the power cord into the proper AC outlet.

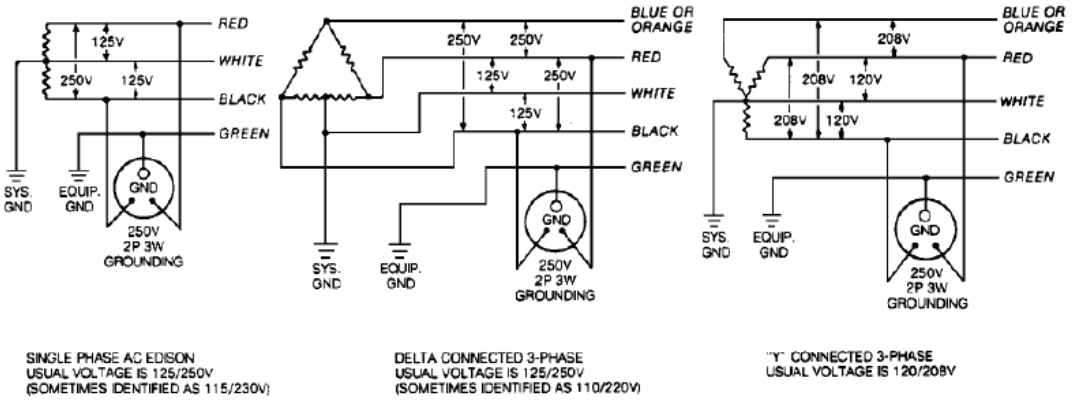
2XX Volt Version Fast Heating Rates

CAUTION

The Wiring of the Plug and Receptacle in the wall are the responsibility of the Customer.

Recommendations follow below.

Typical Voltage Configuration US Voltage Supplies



Receptacle terminal identifications are according to NEMA standards 5/18/64

If the customer has a 208 Volt Line Voltage Supply

1. The Fast heating performance will probably be degraded.
2. Request that the customer have his electrician rewire the Line to 240 Volts.
3. The Chromatograph Heater must be matched to the appropriate supply voltage.
4. The Supply voltage is not typically guaranteed by the Power Supplier.
5. The Supplier only expects the Line Frequency to be within their specifications.

The Clarus 600/680/690 GC systems are all provided without a plug on the line cord. A 15 amp plug is to be supplied by the customer to match the receptacle in their facility. If the Customer is asking for a recommendation of Plug and Socket the following may be recommended.

The following information is for assistance to the Customers' electrician:

- Typical colors of wiring are listed in the tables below.
- The Line cord all of the 2XX units shipped from the factory are wired per the UK standard of:
- Brown, Blue and Green with Yellow Stripe.
- For 2XX Volt installations the Brown and the Blue Wires constitute the Mains voltage.
- The Green with Yellow stripe wire is the Protective earth or Ground wire.

Standard wire colors for plugs			
Region	**Live**	**Neutral**	**Protective earth**
EU,UK, Australia & South Africa (IEC 60446) 240 Volt	brown	blue	Green/yellow
UK, Ireland & Australia before 1969 240 Volt	red	black	green
United States and Canada 120 Volt (*screw color*)	black (*brass*)	white (*silver*)	green (*green*)

Standard Wire Colors for Wall Sockets

Region	Live	Neutral	Protective earth
EU (IEC 60446) 2XX Sockets	brown or black	blue	Green/yellow

Note: The colors above represent current European practice but older installations may be to previous national standards.

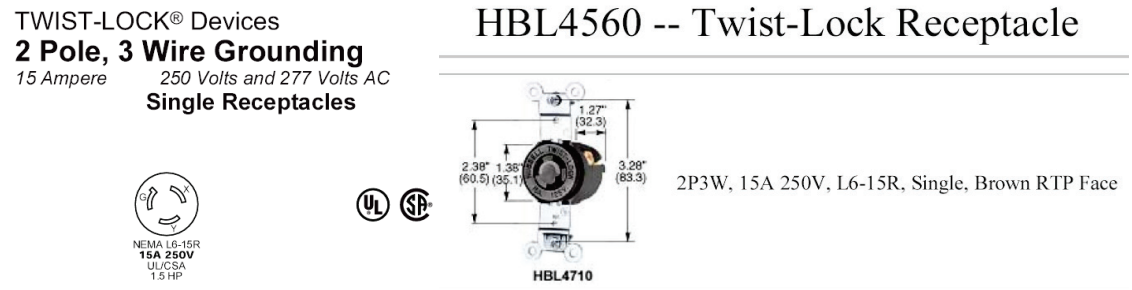
UK & Australia 2XX sockets	brown or red (on older installs)	blue or black (on older installs)	green/yellow (core is usually bare and should be sleeved at terminations)
United States and Canada 120 Volt sockets	black (or red)	white	green or bare

Recommended Line Plug for the Clarus 600/680/690 GC
(U. S.A. and Canada)

PerkinElmer recommended Plug for the Line cord on a 2XX Volt system in the United States. Other Countries must follow the Standards of the Country of installation.



For the United States and Canada the component is available from PerkinElmer under P/N 09290241.



For the United States and Canada this receptacle is available as P/N 09290242.



WARNING

PerkinElmer Service Engineers are not allowed to wire the wall receptacles at the customer site.

It is required that the Proper Electrical and fire Codes in force at the account be observed in the wiring process of the wall receptacles.

For all Chromatograph systems, the electrical supply should contain a "slow blow" circuit breaker capable of handling 300% of the rated current for periods of 3 seconds. Also, the Gas Chromatograph, computer and other accessories should all be connected to the same electrical ground, and the power supply should be free of transients in excess of 40 V peak to peak over the nominal voltage.

***NOTE**: Make sure the Line Voltage matches the Heater that is installed in the GC.*

There will be a tag on the heater wires as they exit the rear of the oven which contains the part number of the heater.

For part numbers see the table below.

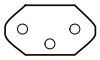
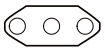
Heater Selection Table

Power Specification: Heater Part Numbers **Standard Heat Rate Heaters** N650-5150 (120 Volt / 20 Amp) Same as the Clarus 500 N650-5150 (230 Volt / 10 Amp) Same as the Clarus 500 **Fast Heat Rate Heaters** N650-5152 (220 Volt) N650-5153 (230 Volt) N650-5154 (240 Volt)	All electrical supplies must be smooth, clean, and free of line transients greater than 40 V peak to peak and must meet and remain within the following tolerances: **For a GC with slow heating rates as standard** 120 VAC ±10% @ 50/60 Hz ±1% @ 20 Amps, 2400 VA Maximum. 230 VAC ±10% @ 50/60 Hz ±1% @ 10 Amps, 2400 VA maximum. **For GC with Optional heater for fast heating rate;** 220VAC ±5% @ 50/60 Hz ±1% @ 3120 VA Maximum 230VAC ±5% @ 50/60 Hz ±1% @ 3120 VA Maximum 240VAC ±5% @ 50/60 Hz ±1% @ 3120 VA Maximum Instruments and peripherals should not be connected to circuits with large inductive or large and frequent loads (i.e., large motors, discharge lamps, photocopy systems, radio transmitters, etc.).

Connecting the Clarus 600/680/690 GC to Line Power Outside the U.S.

To connect the Clarus 600/680/690 GC to line power outside the U.S., connect a properly grounded plug to the power cord and change the wiring on the AC Distribution P.C. board as described in the following procedures.

Table 3- 2 shows the available plugs, their electrical power, and the countries where they are used.

 *13A, 250V Two-Pole with Earth Contact for use in Great Britain, Ireland, and Hong Kong*	 *10/16A, 250V Two-Pole with Earth Contact for use in Austria, Germany, Netherlands, Norway, Portugal, and Sweden*
 *10/16A, 250V Two-Pole with Dual Earth Contacts for use in Austria, Belgium, France, Germany, Netherlands, Norway, Portugal, Spain, Sweden, and Finland*	 *10A, 250V Two-Pole with Earth Contact for use in Switzerland*
 *10A, 250V Two-Pole with Earth Contact for use in Denmark*	SlowHeating Only  *20A,120V for use in the United States*
 *10A, 250V Two-Pole with Earth Contact for use in Australia, New Zealand, New Guinea, and China*	 *10A, 250V Single Phase for use in South Africa, India, and Hong Kong*
 *10A, 250VAC for use in Italy and North Africa*	 *10A, 250V for use in Israel*

In Countries with 230V Single-Phase Power



WARNING

Only a PerkinElmer Service representative or similarly trained and authorized personnel should ever change or replace the plug for the Clarus 600/680/690 GC.

1. Select the appropriate plug and connect it to the three power cord wires as follows:
   - Connect the brown wire to line (AC).
   - Connect the blue wire to neutral (ACC).
   - Connect the green/yellow wire to chassis ground.
2. Remove the three screws that secure the left side panel to the Clarus 600/680/690 GC. Then remove the panel.

3. Verify that the brown wire from the line filter is connected to E1, that the blue wire from the line filter is connected to E2, and that the yellow/green wire is connected to E3 on the AC Distribution P.C. board (see Figure 3-4).

4. Make sure that a 15-A Slow-Blow fuse is installed in the F3 fuse holder.

5. Replace the left side panel and connect the power cord to a suitable electrical outlet.

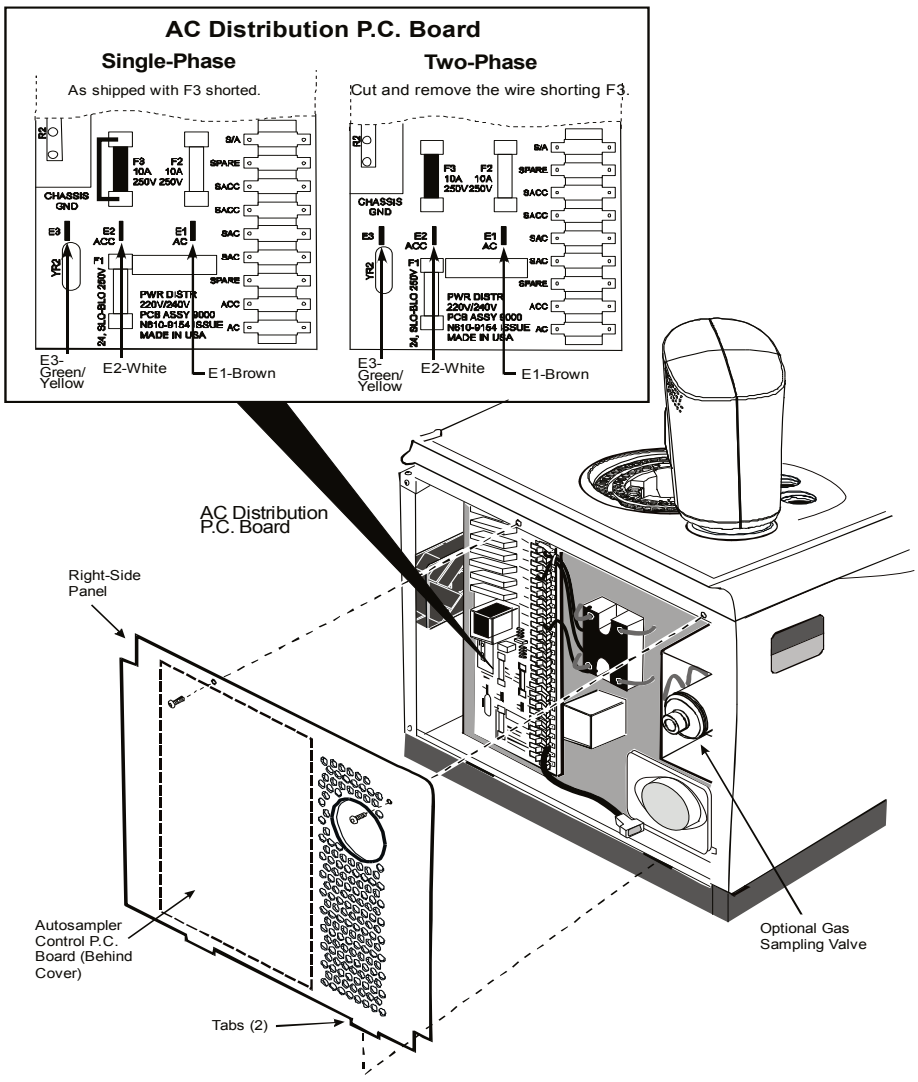
In Countries with 230V Two-Phase Power



WARNING

Only a PerkinElmer Service representative or similarly trained and authorized personnel should ever change or replace the plug for the Clarus 600/680/690 GC.

1. Select the appropriate plug and connect it to the three power cord wires as follows:
   - Connect the brown wire to line (AC Hot).
   - Connect the blue wire to line (AC Hot).
   - Connect the green/yellow wire to chassis ground.
2. Remove the three screws that secure the Left side panel to the Clarus 600/680/690 GC and remove the panel.
3. Verify that the brown wire from the line filter is connected to E1, that the blue wire from the line filter is connected to E2, and that the green/yellow wire is connected to E3 on the AC Distribution P.C. board (see below).
4. Cut and remove the wire that is used as a jumper across the F3 fuse holder. Make sure that the 10-A Slow-Blow fuse remains installed in the F3 fuse holder.
5. Replace the left side panel and connect the power cord to a suitable electrical outlet.



AC Distribution P.C. board connections for countries that use single-phase or two-phase power.

Checking Heater Selection



WARNING

The Clarus 600/680/690 GC Fast heating specifications will not be met if the heater in the chromatograph oven does not match the Line or mains voltage form the wall. Your PerkinElmer service representative must verify that the heater matches the mains supply before completing the installation process.

Preparing the Laboratory

The following sections describe the laboratory requirements in detail for the Clarus GC.

Environmental Conditions

The laboratory in which the Clarus GC is located must meet the following conditions:

- A corrosive-free environment.
- The instrument will operate with a laboratory temperature between 10 and 35 °C (50 - 95 °F). For optimum instrument preformance, the room temperature should be controlled at 20° ± 2 °C.
- The environment should be relatively dust-free to avoid sample and instrument contamination problems.
- Free of excessive vibration.
- Before the instrument is installed, the area around, under and behind the instrument's planned location is to be cleared of any dirt and dust to prevent their entry into the instrument's interior which could cause a negative effect on performance.

The Clarus GC has been designed for indoor use. Do not use the instrument in an area where explosion hazards may exist.

Exhaust Vent Requirements

Exhaust venting is important for the following reasons.

- It protects laboratory personnel from toxic vapors that may be produced by some samples.
- It helps to protect the instrument from corrosive vapors that may originate from the sample(s).
- It removes dissipated heat produced by the instrument and power supply.



WARNING

The use of Clarus GC without adequate ventilation to outside air may constitute a health hazard.

NOTE: *Local electrical codes do not allow PerkinElmer Service Engineers to install the blower and vent assembly.*

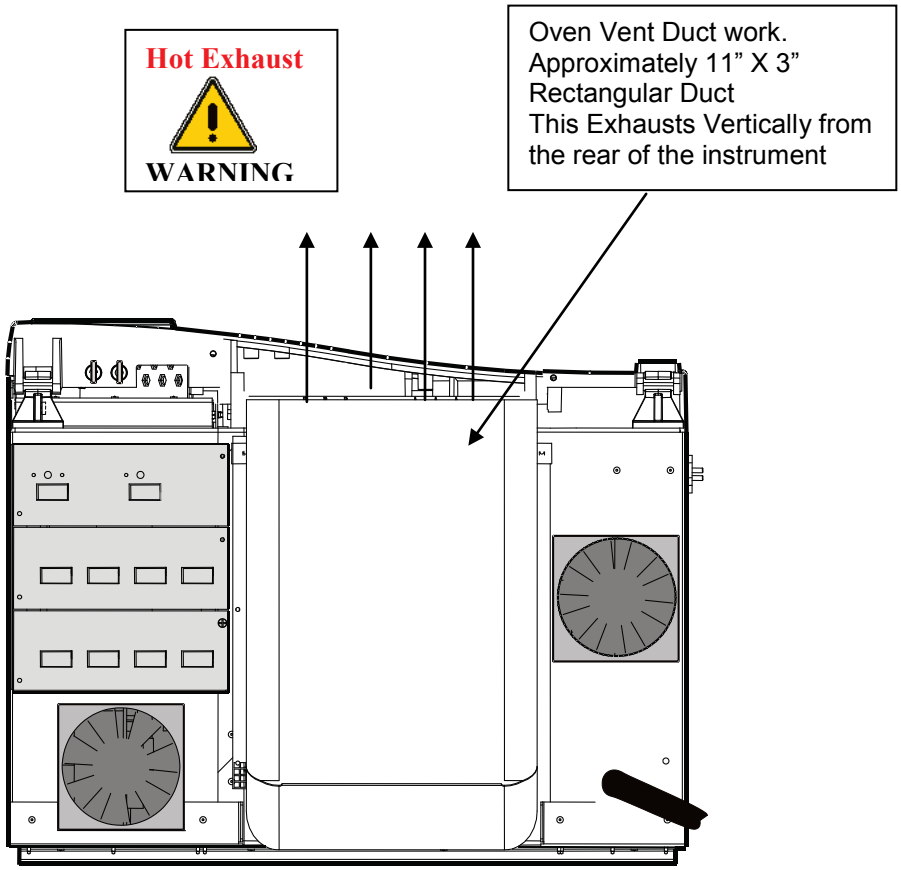
The blower capacity depends on the duct length and number of elbows or bends used to install the system. If an excessively long duct system or a system with many bends is used, a stronger blower may be necessary to provide sufficient exhaust volume at the instrument.

Alternatively, smooth stainless-steel tubing should be used instead of flexible stainless steel tubing where flexibility is not required to reduce system friction loss or "drag." If smooth stainless steel is used, there must be a way to move the vent hood out of the way for servicing. A length of smooth stainless steel ducting has 20-30% less friction loss than a comparable length of flexible ducting. When smooth stainless steel tubing is used, elbows must be used to turn corners. These elbows should turn at a centerline radius

of 150mm with a maximum bend angle of 45 degrees to reduce friction losses, and the number of elbows should be minimized.

Additional recommendations on the venting system include the following items.

- Make sure the duct casting is installed using fireproof construction. Route ducts away from sprinkler heads.
- The duct casing and venting system should be made of materials suitable for temperatures greater than 70 °C (158 °F). It should be installed to meet local building code requirements.
- Locate the blower as close to the discharge outlet as possible. All joints on the discharge side should be airtight, especially if toxic vapors are being carried.
- Equip the outlet end of the system with a back draft damper and take the necessary precautions to keep the exhaust outlet away from open windows or inlet vents. In addition, extend it above the roof of the building for proper dispersal of the exhaust.
- Equip the exhaust end of the system with an exhaust stack to improve the overall efficiency of the system.
- Make sure the length of the duct that enters into the blower is a straight length at least ten times the duct diameter. An elbow entrance into the blower inlet causes a loss in efficiency.
- Provide make-up air in the same quantity as is exhausted by the system. An "airtight" lab will cause an efficiency loss in the exhaust system.
- Ensure that the system is drawing properly by using an air flow meter.
- Equip the blower with a pilot light located near the instrument to indicate to the operator when the blower is on.



NOTE: *Local electrical codes / fire codes **do not allow PerkinElmer Service Engineers** to install the blower and vent assembly.*

The blower capacity depends on the duct length and number of elbows or bends used to install the system. If an excessively long duct system or a system with many bends is used, a stronger blower may be necessary to provide sufficient exhaust volume at the instrument.

Alternatively, smooth stainless-steel tubing should be used instead of flexible stainless steel tubing where flexibility is not required to reduce system friction loss or "drag." If smooth stainless steel is used, there must be a way to move the vent hood out of the way for servicing. A length of smooth stainless steel ducting has 20-30% less friction loss than a comparable length of flexible ducting. When smooth stainless steel tubing is used, elbows must be used to turn corners. These elbows should turn at a centerline radius of 150mm with a maximum bend angle of 45 degrees to reduce friction losses, and the number of elbows should be minimized.

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- Ensure that the system is drawing properly by using an air flow meter.
- Equip the blower with a pilot light located near the instrument to indicate to the operator when the blower is on.

Clarus 600/680/690 GC Requirements

Laboratory Space Requirements:

Size	
GC with Autosampler:	53 cm x 99 cm x 82cm (21-in. x 26-in. x 32-in.)
Weight	
GC:	64.5 kg (142 lb)
Autosampler:	4.5 kg (10 lb)
Physical Configuration:	Single unit for use on standard laboratory bench which can be interfaced to a computer.
Bench Space:	The laboratory bench should be sturdy enough to support the full weight of the GC as well as additional equipment (i.e., computer). Expect the total weight of the GC and accessory equipment to be at least 91 kg (200 lb). Allow a minimum clearance of 10.2 cm (4 in.) on each side, 15.2 cm (6 in.) at the rear of the GC, and 137.2 cm (54 in.) at the top of the GC. If this is not possible, install the GC on a bench that has wheels. Do not position the equipment so that it is difficult to operate the AC power switch.
Peripherals, Printers, etc.	Allow at least 61 cm (24 in.) on either side of the GC to accommodate additional equipment.

Environmental Requirements:

Pollution Degree:	This instrument will operate safely in environments that contain nonconductive foreign matter up to Pollution Degree 2 as defined in IEC 61010-1.
Laboratory Environment:	Install the GC in an indoor laboratory environment that is clean and is free of drafts, direct sunlight and vibration. The laboratory should be free of flammable, explosive, toxic, caustic, or corrosive vapors or gases and should be relatively free of dust. The ambient laboratory temperature should be between 10 °C and 35 °C (50 °F and 95 °F) with a relative humidity between 20% and 80% with no condensation. The GC will operate safely between 5 °C and 40 °C (41 °F and 104 °F).

Electrical Power Requirements:

Power Consumption:	2400 VA (volt-amps) for the GC.
Power Specification:	All electrical supplies must be smooth, clean, and free of line transients greater than 40 V peak to peak and must meet and remain within the following tolerances: 120 VAC ±10% @ 50/60 Hz ±1% 230 VAC ±10% @ 50/60 Hz ±1% Instruments and peripherals should not be connected to circuits with large inductive or large and frequent loads (i.e., large motors, discharge lamps, photocopy systems, radio transmitters, etc.).
Power Outlets:	A minimum of one dedicated 120 VAC outlet at 20 A or one 230 VAC outlet at 10 A (minimum) is required for the GC.

Safety Requirements:

Gas Cylinders and Gas Delivery Lines:	All gas cylinders should be firmly clamped to a suitable surface. Care must be taken not to kink or overstress the gas delivery lines.
Hydrogen:	Ensure that all hydrogen lines and connections are leak-free. When using a hydrogen tank, install an in-line hydrogen snubber (P/N 00090038) between the tank regulator and the delivery tubing.
Ventilation:	Always provide adequate ventilation. When analyzing hazardous compounds, such as pesticides, it may be necessary to arrange for venting the detector effluent into a fume hood.

Gas Requirements:

All gases must have a minimum purity of 99.995%. Gases used with the mass spectrometer and ELCD detectors require a minimum purity of 99.999%. Gas cylinders should be located outside of the laboratory whenever possible and should always be stored and operated in the vertical position. Always use copper tubing that is free of grease, oil, and organic material for all gases delivered to the Clarus GC.

Helium, Nitrogen, 8.5% H_2/91.5% Helium 95% Argon/5% Methane:	A number 1A (200 ft^3) gas cylinder should be used for all carrier gases. Filter all gases (except methane) through a moisture filter and/or hydrocarbon trap and de-oxo filter. Argon/methane should be filtered through a moisture filter and a de-oxo filter. Gas delivery pressure to the GC should be 60 – 90 psig (414-621 kPa).
Air:	A number 1A (200 ft^3) gas cylinder of compressed air or an air compressor can be used. All air should be filtered through a moisture filter. Do ***NOT*** use "Breathing Air." When using manual pneumatics, gas delivery pressure to the GC must not exceed 30 psi (207 kPa). If this is not possible, secondary regulation will be required. With PPC pneumatics, a delivery pressure range of 70 – 90 psig (483 - 621 kPa) is acceptable.
Hydrogen:	A number 2 (62 ft^3) gas cylinder or a hydrogen generator can be used. All hydrogen should be filtered through a moisture filter. With PPC pneumatics, a delivery pressure range of 60 – 90 (414-621 kPa) psig is acceptable.

Sample Preparation Requirements

Sample preparation is the Customer’s Responsibility

4 Installing the Clarus GC

Install the Autosampler

This section describes how to install the autosampler tower, the autosampler tower cover, and the autosampler syringe. The autosampler tower and the tower cover are shipped together in one box. The autosampler tray is installed in the Clarus GC at the factory.

Remove the Encoder Protector Block

The Clarus GC autosampler has been shipped with an encoder protector block. This is a piece of antistatic polyethylene foam that protects the tower disk during shipment. Before you install the autosampler tower onto your GC, you *must* remove the encoder protector block from the tower disk assembly.

To remove the encoder protector block:

1. Carefully remove the autosampler tower from the shipping box.

2. The piece of foam on top of the tower is the encoder protector block. Carefully slide the block straight out from the tower disk.

CAUTION *Do not twist the block off. This will damage the tower disk.*

Install the Autosampler Tower

Perform the following procedure to install the autosampler tower:

1. The autosampler tower mounting hardware is shipped already secured to the GC top. Remove the three screws and flat washers securing the mounting hardware and use them to secure the tower base to the GC.

2. Position the tower interface harness through the autosampler stop, which is part of the tower base. The harness should be fitted carefully into the hole in the autosampler base assembly (see Note below).

NOTE: *You may find it easier to feed the wires a few at a time. The two yellow tie wraps on this harness indicate the portion of the harness that needs to be fed to the tower. Only one yellow tie wrap should be visible at the autosampler stop.*

3. Install the autosampler tower on the top of the GC as shown in Figure 1. The locating pin on the GC fits into the slotted hole in the autosampler tower base.

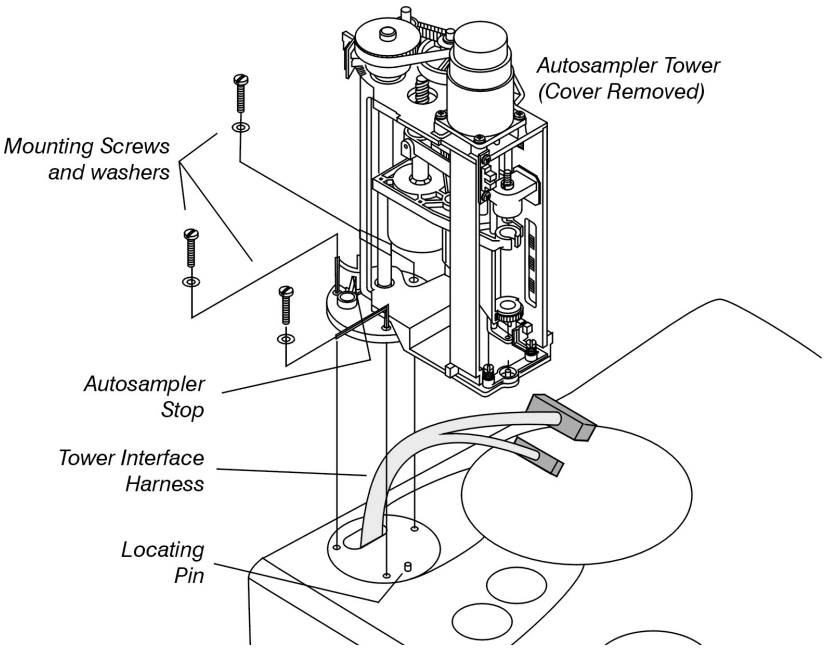


Figure 1. Installing the autosampler tower.

4. Connect the two harness connectors to the tower interface P.C. board. Plug the large connector into J2 and the small connector into J1 (see Figure 2).

These connectors are keyed and connect only one way.

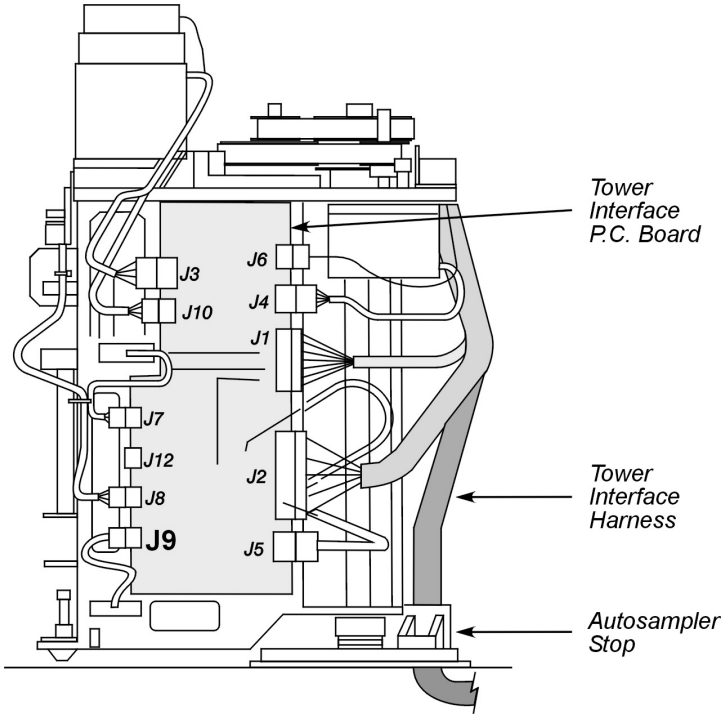


Figure 2. Connecting the tower interface harness to the tower interface p.c. board.

5. Manually rotate the autosampler tower so that the front of the tower (this is where the syringe will be installed) faces the rear of the GC.

6. Secure the harness to the autosampler tower frame using the cable clamp attached to the harness. Attach the cable clamp and the harness ground wire to the underside of the casting as shown in Figure 3, using the screw provided. Make sure the harness is routed correctly before you tighten the screw, or you may have difficulty installing the autosampler tower cover.

NOTE: *There are two holes in the autosampler tower frame casting in the area to which the cable clamp should be secured. Secure the harness cable clamp and ground wire to the inner-most hole.*

7. The screw is already attached to the casting. Use a long Phillips screwdriver to remove the screw, then attach the cable clamp.

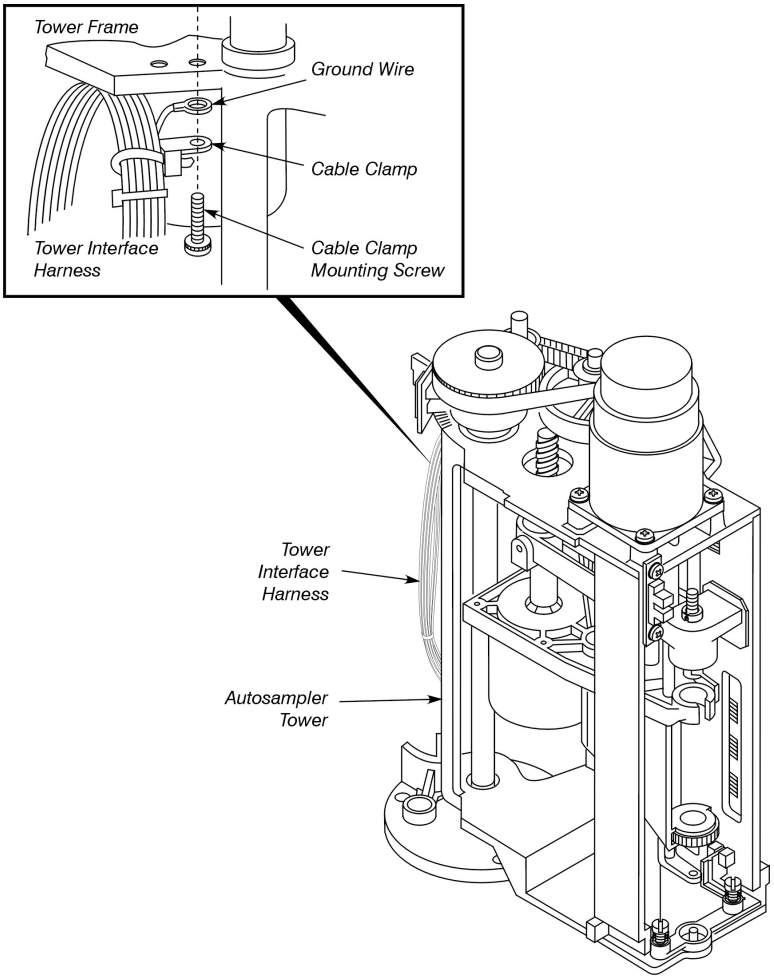


Figure 3. Securing the harness to the tower.

Install the Autosampler Tower Cover

Install the autosampler tower cover by referring to Figures 4 and 5 as you follow this procedure:

1. The tower is shipped with two 3/8-in. long screws installed on the bottom of the tower frame. Loosen but do not remove these screws.

2. Open the cover door and carefully lower the cover onto the tower, aligning the two guides inside the cover with the sides of the tower (see Figure 5).
3. Pull the sides of the cover away from the tower frame just enough to slide the cover tabs onto the two screws.
4. Tighten the two screws. Verify that the cover door opens and closes freely and that it locks when closed. If not, realign the cover until the door locks when closed.

NOTE: *If the door is hitting the door sensor, loosen the two screws that secure the door sensor bracket to the tower, then adjust the sensor up or down until the door no longer hits the sensor (see Figure 4).*

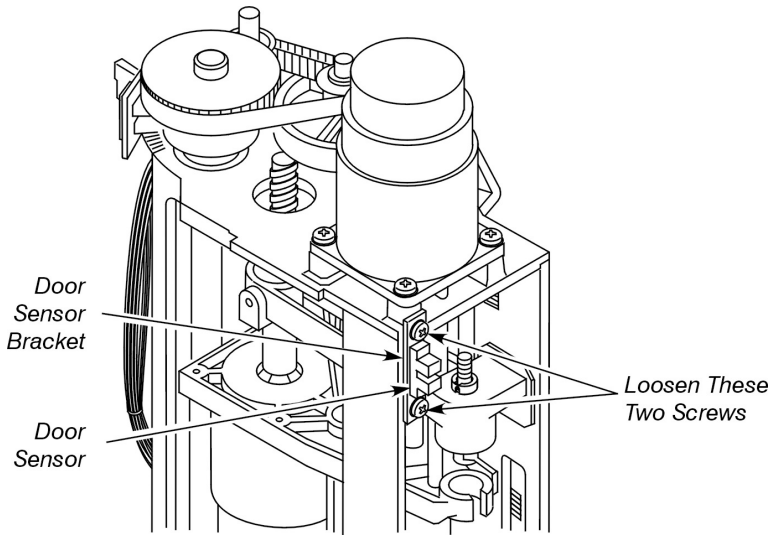


Figure 4. Adjusting the position of the tower door sensor.

Figure 5, on the following page, shows how to install the autosampler tower cover onto the tower.

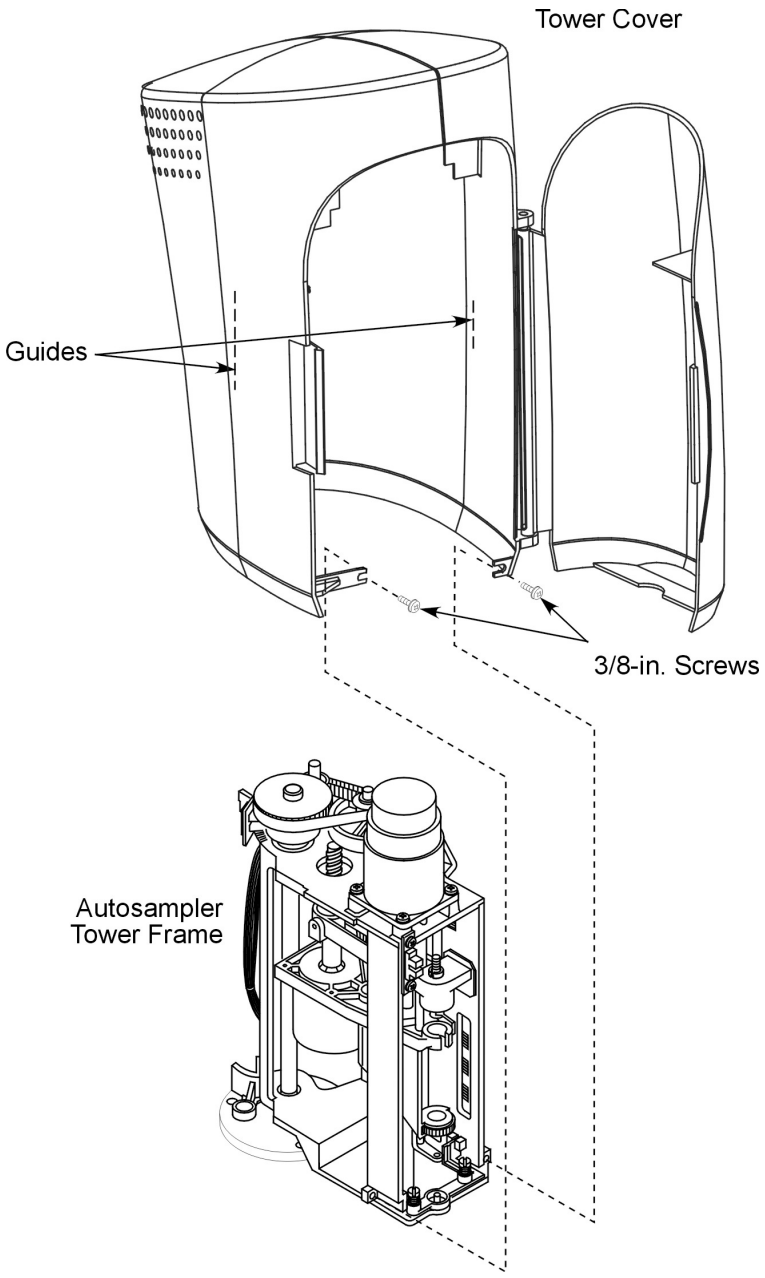


Figure 5. Installing the autosampler tower cover.

Install the Syringe Cover

To install the syringe, proceed as follows:

1. Before you install the syringe, move the autosampler tower so it faces the front of the instrument; this is called the “Park" position (see Figure 6).

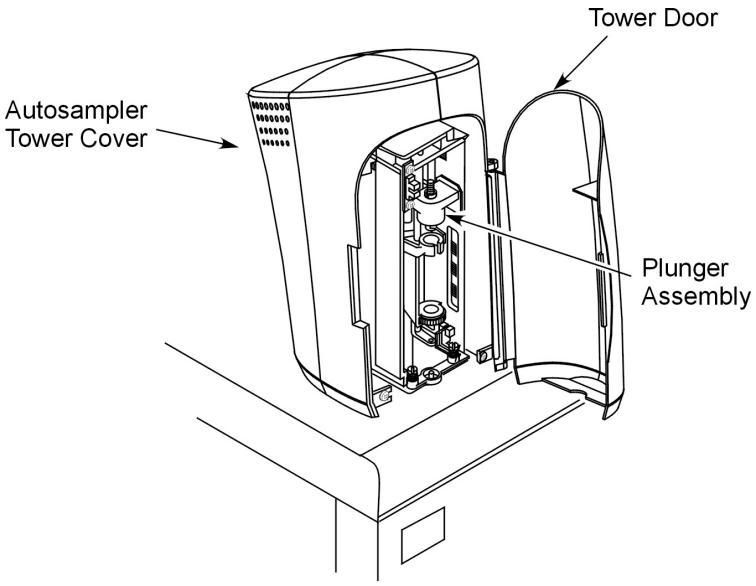


Figure 6. The autosampler tower in the "Park" position.

CAUTION	*Always handle the syringe with extreme care. Be careful not to bend the needle. Before installing the syringe, pull some solvent up into the barrel to moisten the mechanism.*

NOTE: *Use only syringes sold by PerkinElmer for the Clarus GC. Syringe plungers are not interchangeable from one syringe to another.*

2. Lift and turn the plunger cap handle (located on top of the plunger assembly) until the pin locks it in the up position as shown in Figure 7.

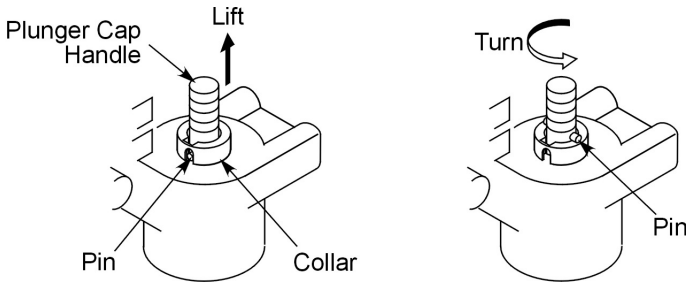


Figure 7. Lifting and turning the plunger cap handle.

3. Insert the syringe needle through the bottom of the carriage assembly and use your finger to carefully guide the needle through the hole in the needle guide (see Figure 8).

CAUTION	*Be careful not to bend the needle when inserting it into the hole in the needle guide.*

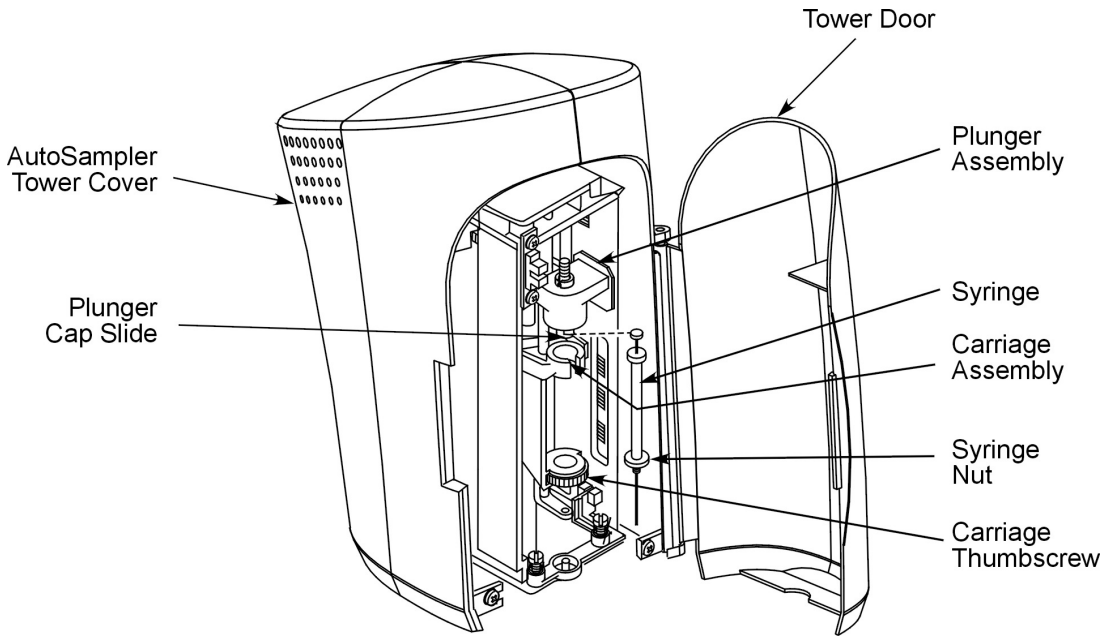
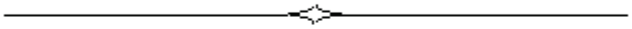


Figure 8. Installing the syringe.

4. Slide the top of the syringe plunger into the plunger cap slide, then lift and turn the plunger cap handle so the pin drops into place. This will lock the syringe plunger into place.
5. Hold the syringe nut with one thumb and, with your other thumb, turn the carriage thumbscrew counterclockwise to tighten it (see Figure 8). ***DO NOT OVERTIGHTEN THE THUMBSCREW***.
6. Carefully guide the needle into the needle locator in the center of the vial locator as you tighten the thumbscrew (see Figure 8).

Your autosampler is now installed.



Connecting the Gas Supply 5



Connect the Gases

The following sections describe how to connect the gas cylinders to the Clarus 600/680/690 GC and how to check the connections for leaks. Use the fittings and copper tubing supplied in the installation kit (P/N 03328000).

Gas Cylinder Connections

Use the following procedure to make the gas cylinder connections:

1. Using a large adjustable wrench (at least 1¼ in.), attach a dual-stage regulator, or two single-stage regulators in series, to each gas cylinder. The regulator outlet fitting must be a 1/8-in. Swagelok type.
2. Connect a 1/4-in. to 1/8-in. adapter to the regulator outlet fitting on all your gas cylinders except the hydrogen cylinder.
3. Insert a hydrogen snubber (P/N 00090038) in the hydrogen cylinder regulator outlet fitting as shown in Figure 9. Use a 1/4-in. to 1/8-in. reducer (P/N 09903212) between the snubber and the regulator.

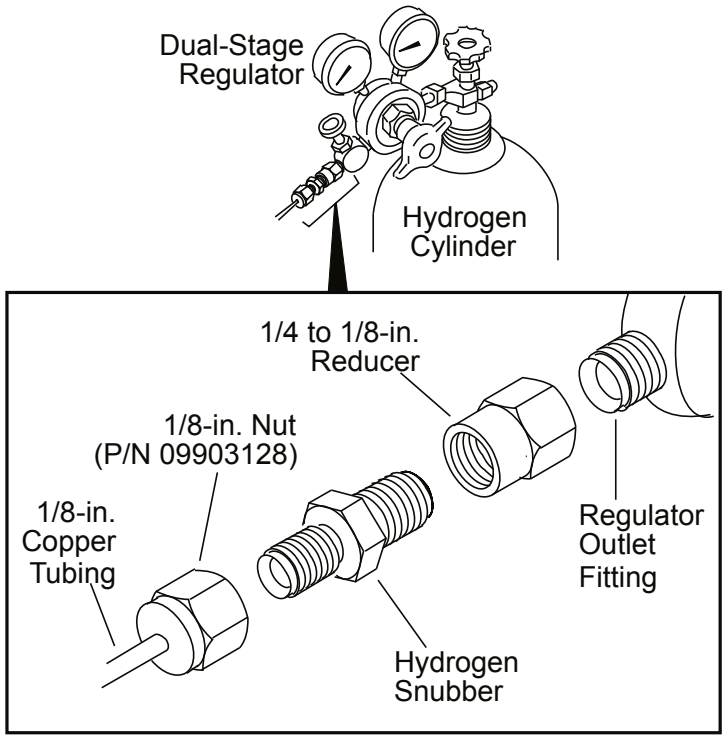


Figure 9. Connecting a hydrogen snubber.

4. Cut a length of 1/8-in. copper tubing long enough to connect a line filter to each gas supply line or gas cylinder.
5. Place a 1/8-in. nut, a 1/8-in. rear ferrule, and a 1/8-in. front ferrule over each end of the copper tubing as shown in Figure 10.

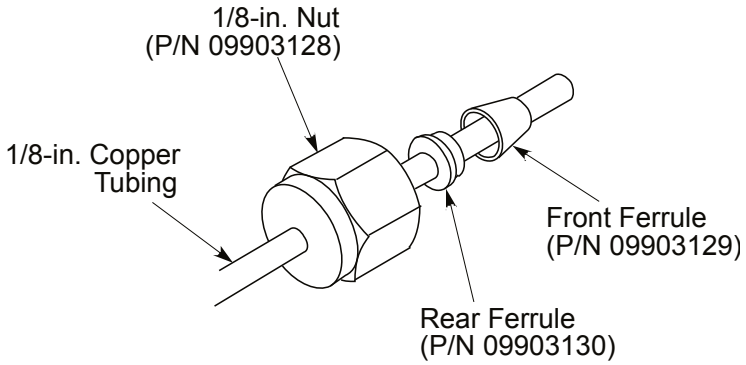


Figure 10. Components of a 1/8-in. tubing connection.

6. Insert one end of the tubing with the nut and ferrules into the end fitting of a suitable line filter (see Table 1), and insert the other end into the regulator outlet fitting or the end fitting of the hydrogen snubber.
7. Insert the tubing into the fitting so that the tubing bottoms firmly, then tighten the nut ½-turn past finger-tight.
8. Cut a piece of 1/8-in. copper tubing long enough to connect the filter to the Clarus GC. To allow for future instrument movement, cut the tubing 2 – 3 ft longer than necessary.

Table 1. Recommended Filters

Detectors	Carrier Gas	Combustion Gas	
		H_2	Air
FID NPD FPD	Hydrocarbon Trap (P/N N9301192) + Oxygen Filter ((P/N N9301179) if capillary column is used	Moisture Filter (i.e., Molecular Sieve) ((P/N N9301193)	Moisture Filter (i.e., Molecular Sieve) ((P/N N9301193)
TCD**	Hydrocarbon Trap ((P/N N9301192) + Oxygen Filter ((P/N N9301179) if capillary column is used		
ECD**	Moisture Filter (i.e., Molecular Sieve) ((P/N N9301193) + Oxygen Filter ((P/N N9301179)		
PID	Hydrocarbon Trap ((P/N N930-1192) + Oxygen Filter ((P/N N9301179) if capillary column is used		

* There is a combination filter system that contains an Oxygen Filter, Oxygen Indicator, and a Moisture Trap. Its part number is N9306002.

** Install a tee fitting on the carrier gas line for the TCD reference line or for the make-up line for the ECD.

Clarus 600/680/690 GC Gas Connections

Perform the following procedure to connect the gases to the Clarus GC:

1. Make a tubing strain relief by forming an approximate 3-in. diameter loop in the tubing so that it is approximately 3 in. from the end of the tubing closest to the Clarus GC as shown in Figure 11. Make this strain relief on all gas lines.

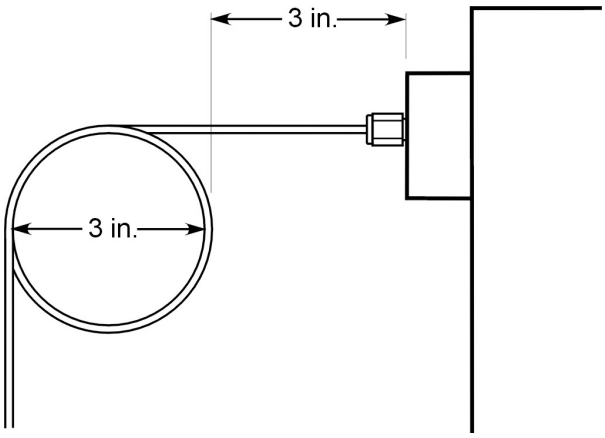


Figure 11. Tubing strain-relief loop.

2. Insert a 1/8-in. nut, a 1/8-in. rear ferrule, and a 1/8-in. front ferrule over the end of each piece of copper tubing as shown in Figure 10.

3. Insert the tubing into the proper fitting on the GC bulkhead for the gas that you are connecting. Be sure the tubing bottoms out firmly in the fitting, then tighten the nut ½-turn past finger-tight.

Figure 12 shows the manual pneumatics bulkhead and PPC bulkheads on the back of the GC. M/U = make-up gas, CARR. = carrier gas, and HYD and H2 = hydrogen gas.

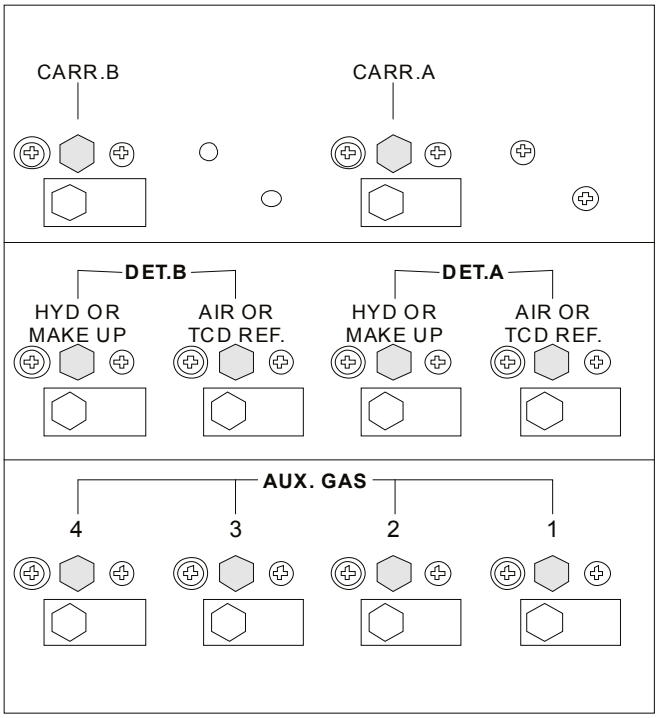


Figure 12. Layout of the Clarus GC PPC bulkheads on the rear panel.

Leak Test the Gas Connections

After connecting the gases to the Clarus GC, check all connections for possible leaks by following this procedure:

1. Set the regulator on the carrier gas cylinder between 60 and 90 psig.

2. Test the carrier gas connections by using either a gas leak-check solution of 50% isopropyl alcohol in water or an electronic leak detector. Check for leaks (indicated by bubbles) at each connection.

CAUTION	*Do **not** use a soap solution to check for gas leaks. It could contaminate the system.*

3. To check a hydrogen or air cylinder for leaks, set the regulator to 30 psig for manual pneumatics and for PPC systems between 60 and 90 psig for H_2 and between 70 and 90 psig for air; test the connections in the same manner that you tested the carrier gas connections.

Common Injector/Detector Gas Connections

Figures 13 through 21 shows how gases are connected to the Clarus 600/680/690 GC in several common injector/detector configurations. Note that the strain relief, which is required on all gas lines, is not shown in these figures.

WARNING

Position incoming gas lines away from the oven exhaust duct so that the hot exhaust will not blow directly on the gas lines.

CAUTION

Thoroughly purge the incoming gas lines before measuring and adjusting the gases. This is especially important when using hydrogen gas.

The following figures show the connections for the injectors and detectors to the PPC bulkheads.

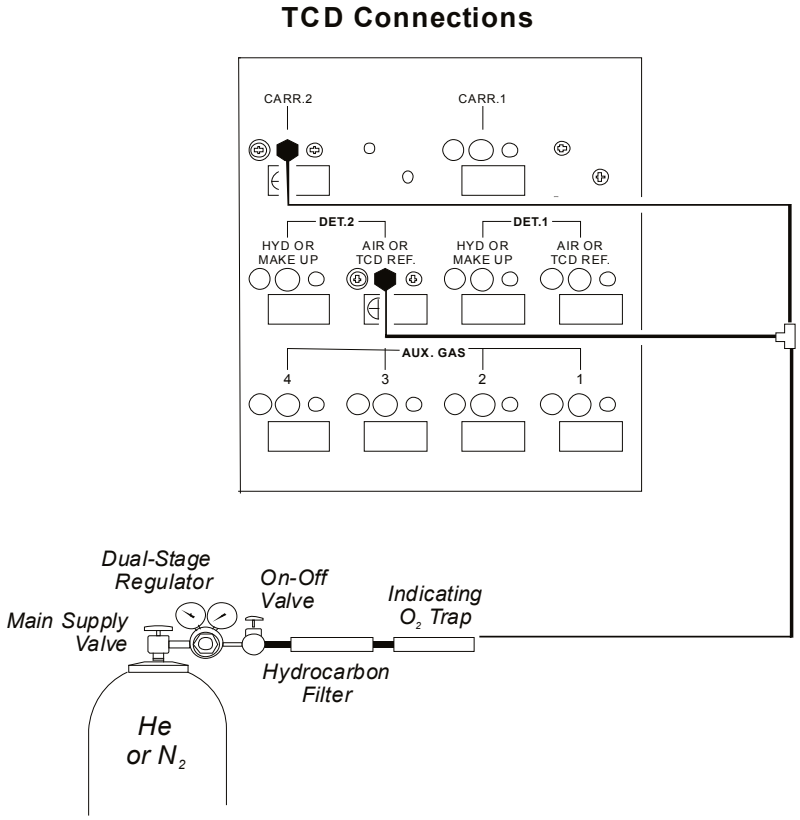


Figure 13. TCD in Channel B (rear channel) with one packed injector.

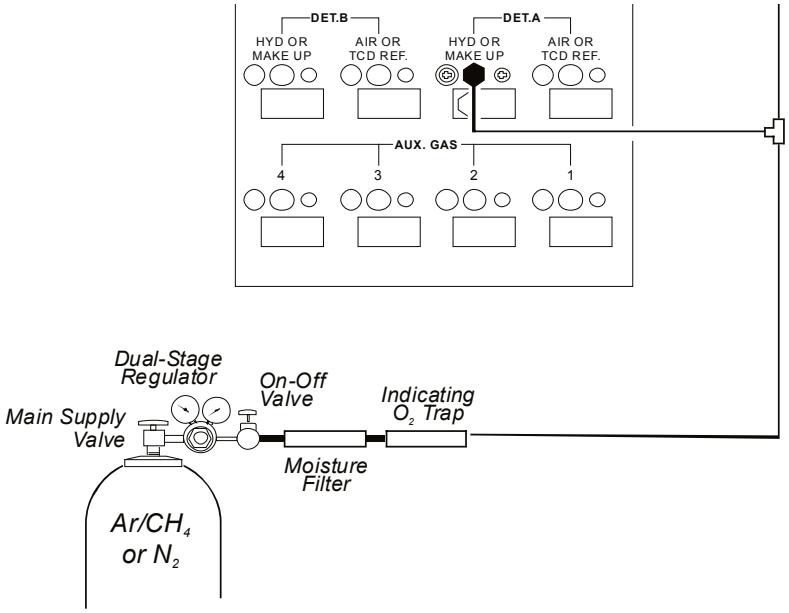


Figure 14. ECD in Channel A (front channel) using argon/methane or nitrogen as the carrier gas, with one injector.

NOTE: *If you are using the ECD with a capillary column at low flows (5 mL/min or less), you can plumb the injector with helium and the make-up line with argon/methane or nitrogen. Helium is the preferred carrier for capillary columns.*

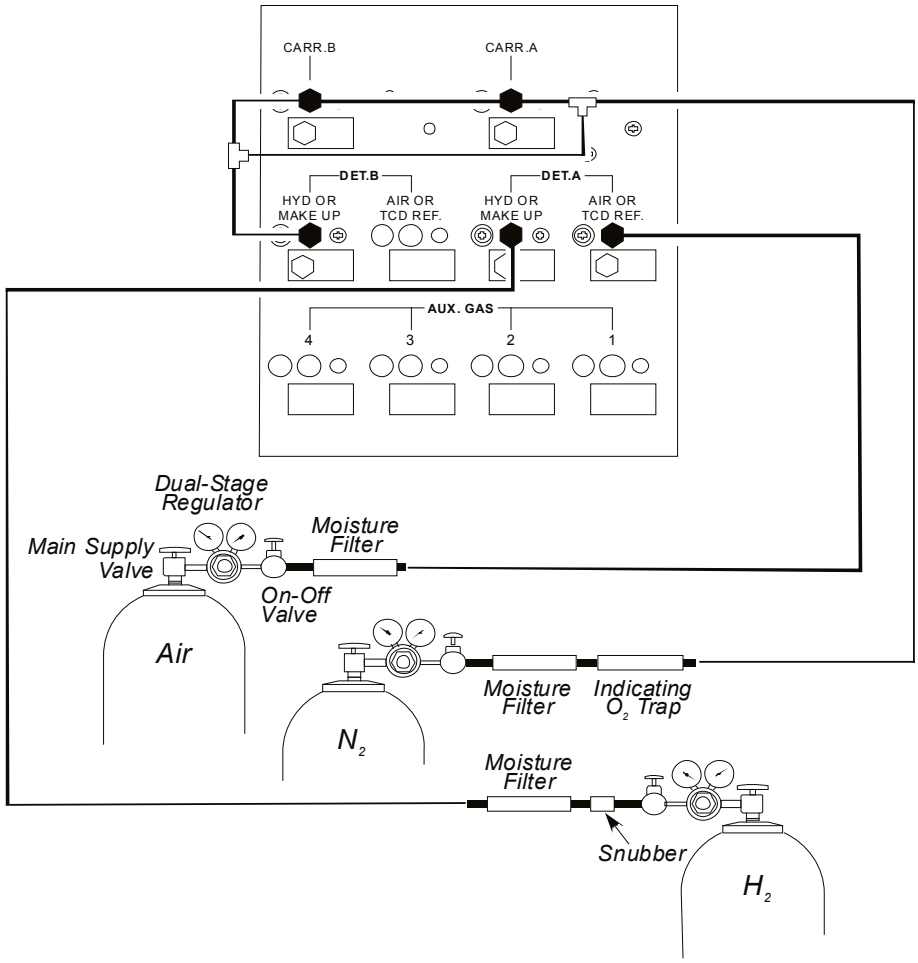


Figure 15. FID, NPD, or FPD in Channel A and ECD in Channel B using nitrogen as the carrier gas, with two injectors.

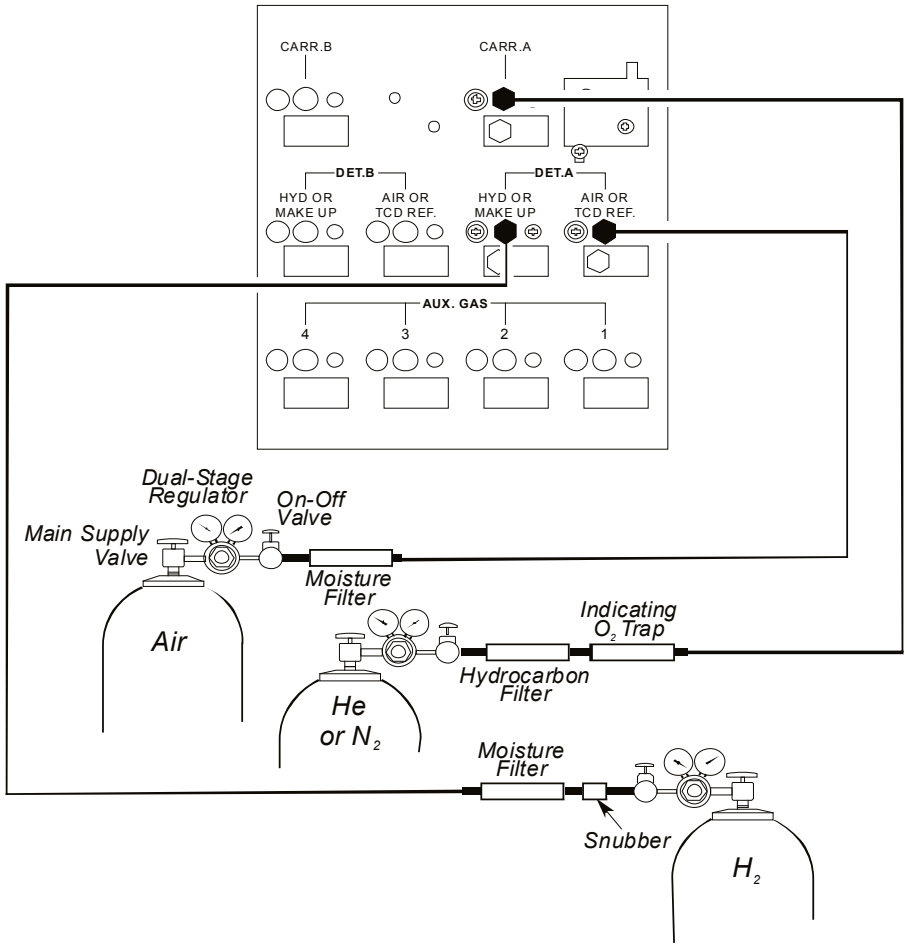


Figure 16. FID, NPD, or FPD in Channel A using helium or nitrogen as the carrier gas, with one injector.

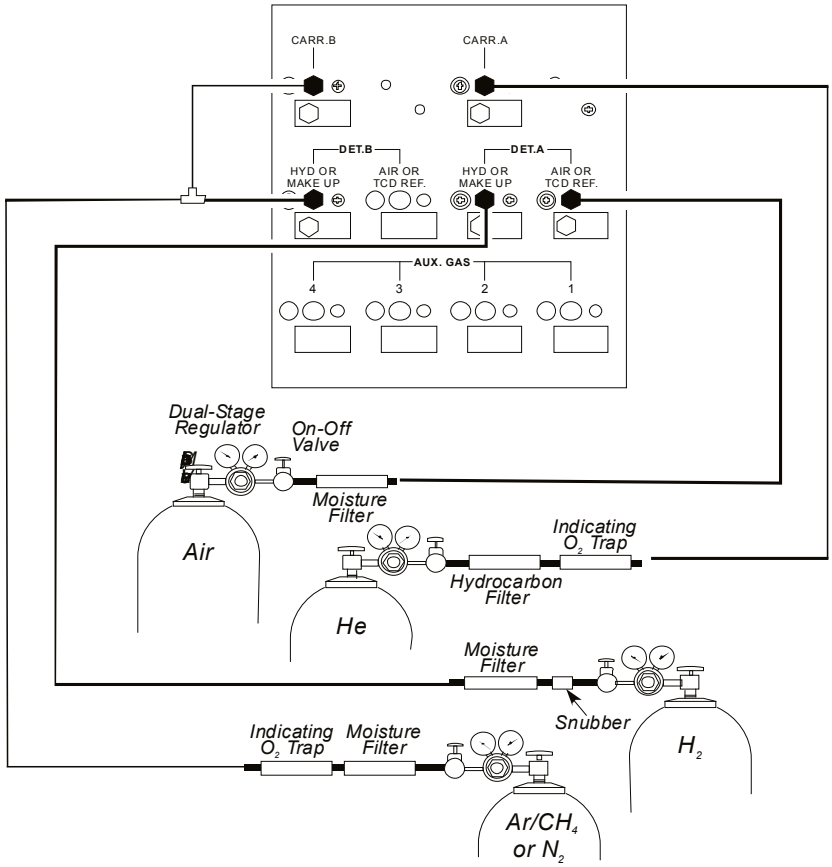


Figure 17. FID, NPD, or FPD in Channel A using helium carrier gas, ECD in Channel B using argon/methane or nitrogen carrier gas, with two injectors.

NOTE: *If you are using the ECD with a capillary column at low flows (5 mL/min or less), you can plumb the injector with helium and the make-up line with argon/methane or nitrogen. Helium is the preferred carrier gas for capillary columns. The FID, NPD, and FPD do not require make-up gas.*

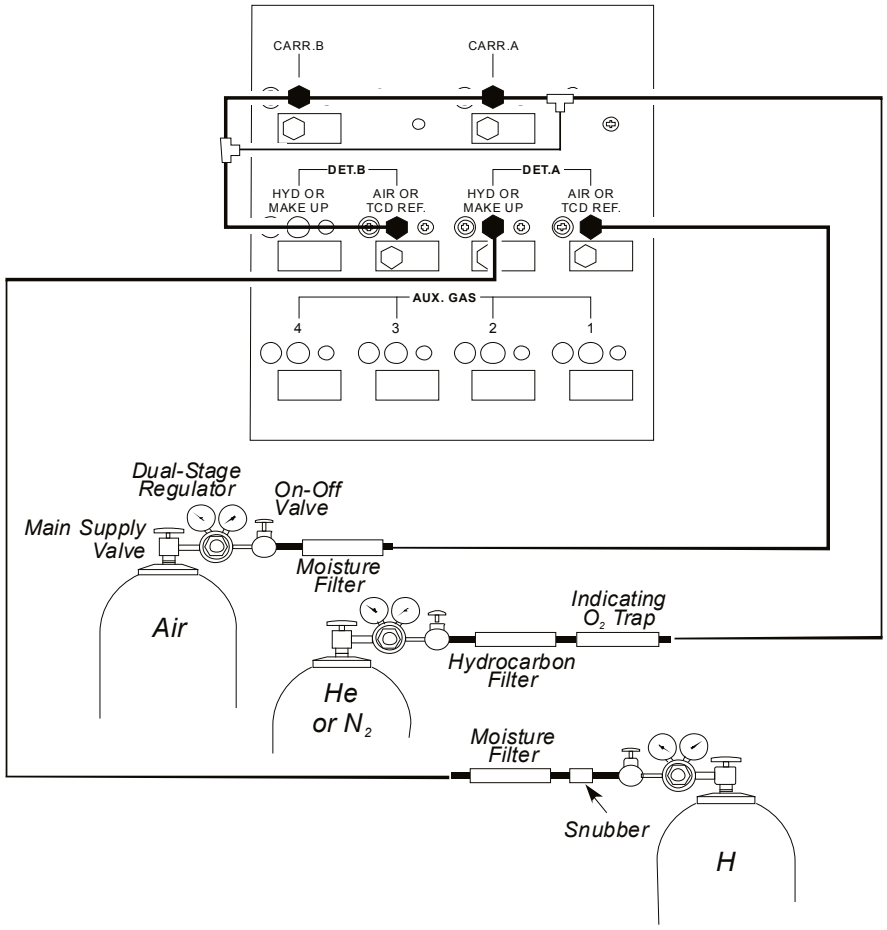


Figure 18. FID, NPD, or FPD in Channel A, TCD in Channel B, with two packed injectors.

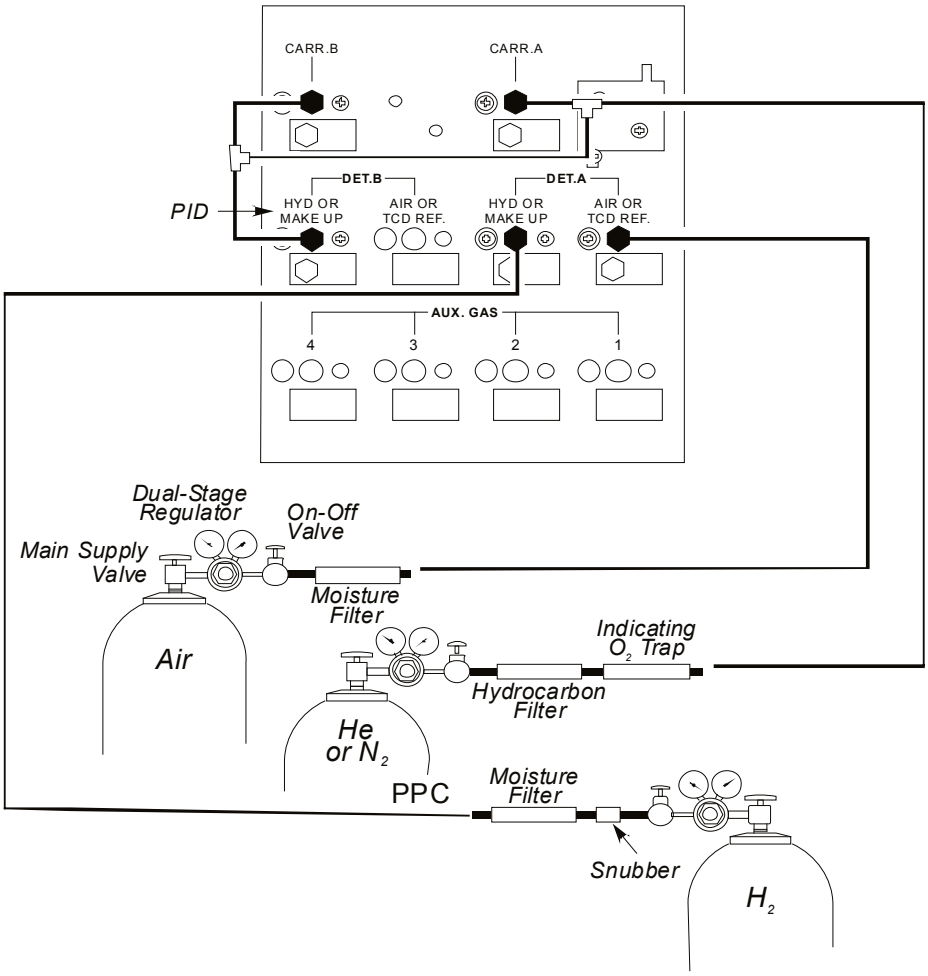


Figure 19. FID, NPD, or FPD in Channel A, PID in Channel B, with two packed injectors.

Connecting the Accessories 6

Install the NPD Bead Assembly

If you purchased your Clarus GC equipped with a Nitrogen Phosphorus Detector (NPD), the NPD bead assembly is packaged separately and shipped inside the GC oven. To install the NPD bead assembly, use the following procedure as you refer to Figure 20.

CAUTION	*After installing a NPD bead, you must set up the NPD and condition a new bead before you can use it. Refer to Setting Up a NPD in Chapter 8 of the Clarus 600/680/690 GC User's Manual (P/N 09936780).*

1. Open the detector cover.
2. Remove the bead assembly package from the Clarus GC oven.
3. Loosen the knurled ring on the NPD collector assembly. Then remove the collector assembly.

CAUTION	*You may find it easier to first remove the coaxial cable from the collector assembly before you remove the collector assembly.*

4. Remove the screw that secures the bead transformer assembly to the top of the Clarus GC oven.
5. Plug the bead assembly (P/N N6120092) into the connector on the bead transformer. The connector is keyed so that the bead assembly can only be inserted one way.
6. Carefully insert the bead portion of the bead assembly in the detector body as shown in Figure 20.
7. Secure the bead transformer to the top of the oven with the screw removed in step 4.
8. Replace the collector assembly on the detector body and secure it by tightening the knurled ring. If the coaxial cable was removed, connect it to the collector assembly.

CAUTION	*Ensure that the polarizing wire is still connected to the polarizing pin on the detector body before you set up and run the NPD.*

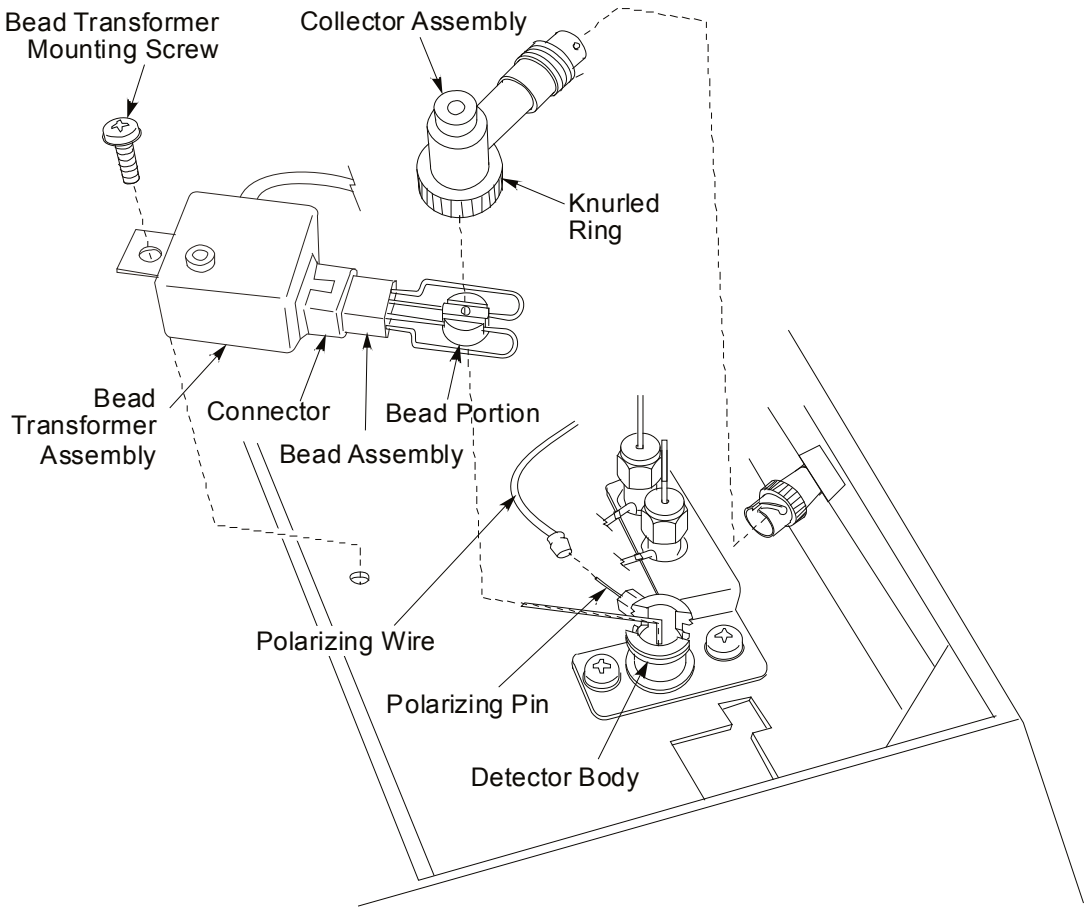


Figure 20. Exploded view of the Nitrogen Phosphorus Detector.

Connect a Recorder or Integrator

This section describes how to connect an optional recorder or integrator to the Clarus GC. To connect either one of these devices, use the following procedure.

1. To connect a device to amplifier Channel A or B, first configure output 1 or 2 for a recorder or integrator as described in Chapter 3, Initial Setup Procedures, in the *Clarus 600/680/690 GC User's Manual*.

2. Open the small access cover on top of the Clarus GC. With a large flat bladed screw driver loosen the large screw. On the main cover is a screw well. Loosen this screw with the large flat bladed screwdriver. You will then be able to lift the entire cover up and have access to the recorder or integrator.

3. To connect a recorder or integrator to Channel A, connect the ground lead of the device to TB1-13 and the positive lead of the device to TB1-14. Make sure you route the wires around the strain relief posts as shown in Figure 22.

4. To connect a recorder or integrator to Channel B, connect the ground lead to TB1-15 and the positive lead to TB1-16.

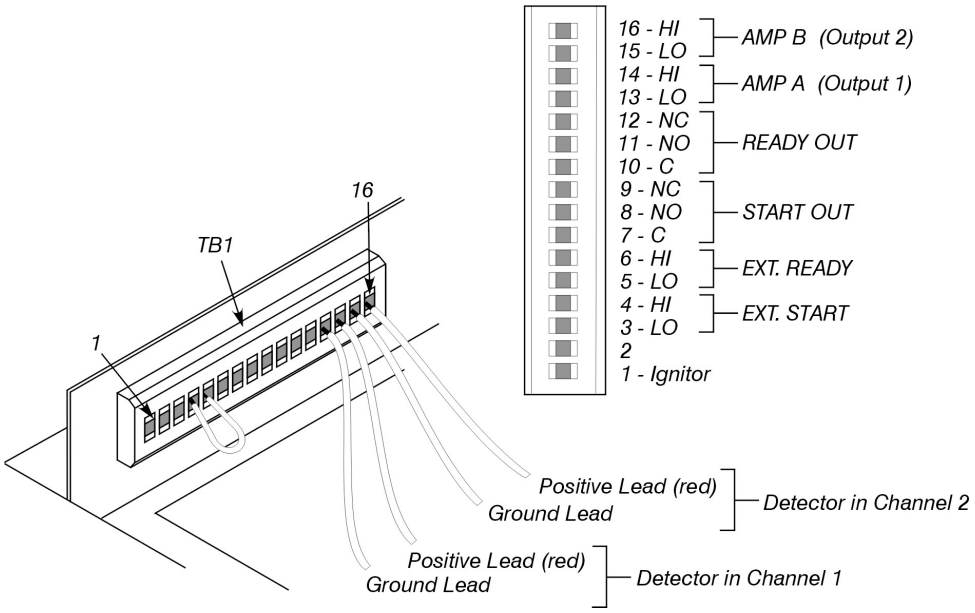


Figure 21. Recorder and integrator connections at TB1.

In addition to the amplifier Channel A and B contacts, TB1 has READY and START contacts. Table 3 describes the functions of these contacts.

Table 3. READY and START Connections at TB1

Connection	Function
READY OUT TB1-10 (C), 11 (NO), 12 (NC)	*Instrument READY OUT Relay:* These contacts are used to tell an external device that the Clarus GC is ready. The normally open contact (NO) is closed in the Ready state.
START OUT TB1-7 (C), 8 (NO), 9 (NC)	*Instrument START OUT Relay:* These contacts are used to start an external device, such as an integrator, when the Clarus GC starts a chromatographic run. The normally open contact is closed for 1 sec when a run is started.
EXT. READY (TB1-5, 6) TB1-6 is signal TB1-5 is ground	*External Ready In:* The Clarus GC requires that these contacts be shorted together to become ready and is shipped with a link across them. When using an external device, such as an integrator, remove the link and wire the device to provide a contact closure indicating the Ready state. This will prevent the instrument from becoming ready before the external device is ready.
EXT. START	*External Start In:* Shorting these contacts will cause

(TB1-3, 4) TB1-4 is signal TB1-3 is ground	the Clarus GC to start a chromatographic run. It is equivalent to pressing the RUN key.

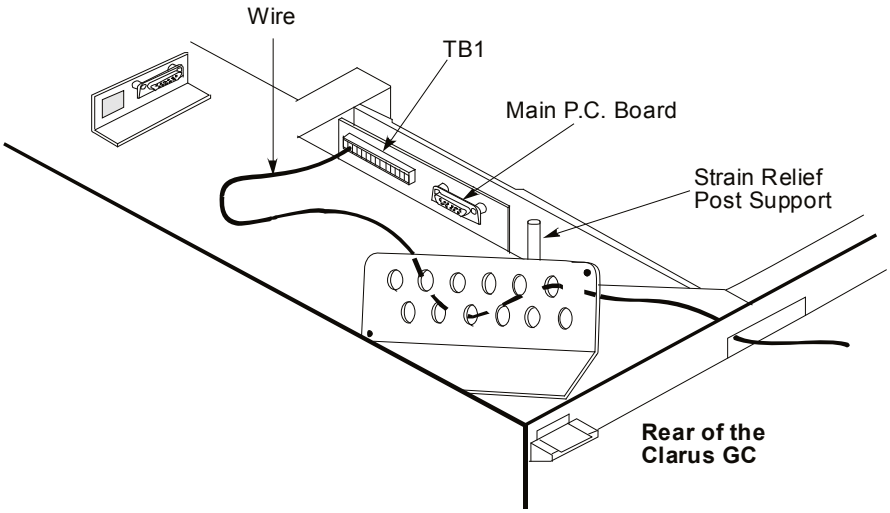


Figure 22. Routing recorder/integrator wires around the strain relief posts.

PPC Restrictor Information 7

Available Restrictors

Restrictor Number	**Color**	**Helium Flow at 30 PSIG, vent to atmosphere**
1	Silver	1 mL/min
2	Red	3 mL/min
3	Yellow	10 mL/min
4	Black	30 mL/min
5	Green	100 mL/min
6	Blue	300 mL/min

Restrictors for Carrier Gas Control

Injector	**Restrictor Number**	**Helium Flow (mL/min)** (**with a 90 psig inlet pressure and a 10 psig drop across the restrictor)**
Packed	4 (std)	30
Split/Splitless or PSS	6 (std)	300
Split/Splitless or PSS	4 (acc)	30
POC	3 (std)	10
POC	2 (acc)	3

Restrictors for Detector Gas Control

NOTE: *Maintain an inlet pressure between 60 and 90 psig.*

Detector	Gas	Restrictor Number	Nominal Flow Required (mL/min)
FID	Air	6	450
FID	Hydrogen	4	45
NPD	Air	5	100
NPD	Hydrogen	1	2
FPD	Air	5	90
FPD	Hydrogen	4	75
ECD	Argon/Methane	4	30
ECD	Nitrogen	4	30
TCD	Helium	4	30
TCD	Hydrogen	3	30

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